SITE SAFETY PLUS

Construction site health and safety

E: Environment

Published by CITB, Bircham Newton, King's Lynn, Norfolk PE31 6RH

First published in 1982 as GE700 (A4 ring binder x2)
Republished in 2013 as GE700E (A4 book)
Revised 2014, 2015
Reprinted June 2015
Revised 2016
Reprinted Jan 2017
Revised Dec 2017, Dec 2018, Dec 2019, 2022
Revised 2023, 2024

ISBN 978-1-85751-585-5

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Summary of sustainable construction legislation and guidance

This list is not exhaustive and only includes legislation mentioned in this section of GE700.

Legislation and guidance	Enforcement agencies*						
	CIRIA	EA	HSE	LA	NIEA	NRW	SEPA
Acts (primary legislation)							
Climate Change Act 2008		✓			✓	✓	
Climate Change (Scotland) Act 2009							✓
Environmental Protection Act 1990		✓			✓	✓	✓
Regulations (secondary legislation)							
COSHH Regulations 2002			✓				
UK Building Regulations 2010				✓			
Guidance							
ISO 14001 Environmental Management Systems							
ISO 20400 Sustainable Procurement							
ISO 26000 Social Responsibility							
ISO 50001 Energy Management							
Low Carbon Construction Action Plan and Routemap							
Waste Resources Action Programme (WRAP)	✓						
Working with substances hazardous to health: A brief guide to COSHH			✓				
Organisations that have information and guidance on their websites	✓	✓	✓	✓	✓	✓	✓

***Key**

COSHH	Control of substances hazardous to health
CIRIA	Construction Industry Research and Information Association
EA	Environment Agency
HSE	Health and Safety Executive
LA	Local Authorities
NIEA	Northern Ireland Environment Agency
NRW	Natural Resources Wales
SEPA	Scottish Environment Protection Agency

Overview

Environmental pressures are amongst the most serious issues facing the human race. Climate change, greenhouse gases, food production, loss of biodiversity and the harmful effects of plastics and chemicals are constantly in the news. Slowly, the global community is realising the damaging effects of these issues and the accelerating consequences of an increasing world population.

The construction industry can affect the environment in a number of ways. It therefore has a major role to play in protecting natural resources and ensuring that they are passed on to future generations, in good order, for their enjoyment.

This chapter gives a general introduction to the environment. It explains how the environment is defined, what local and global impacts put pressure on the environment and how these link with the overall concept of sustainable development. It also gives a general overview of the legal framework for the regulation of the environment and government environmental targets. It explains how the construction industry is responding to the environmental agenda by using fewer resources, less energy and promoting sustainable development.

1.1 Introduction

Rapid population growth and an ever-increasing demand for resources from economic development are placing huge pressures on our planet. In turn, this has led to an increase in all types of pollution and an acceleration of environmental damage. If everyone in the world lived as we do in Europe, we would need three planets to support us because we consume resources at a much faster rate than the planet can replenish them.

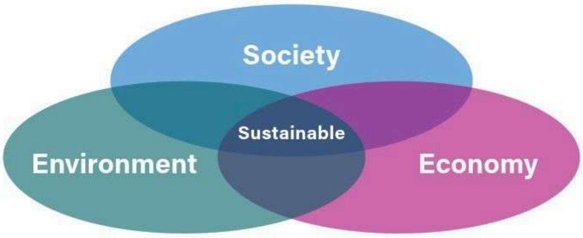
People, consumption, production and the environment are all linked and have a major impact on each other. It is therefore important to consider how construction's contribution to future development can be achieved without causing any further damage. Sustainable living is about respecting the earth's environmental limits.

The terms *sustainability* and *sustainable development* were first established in the paper *Our common future*, released by the Brundtland Commission in 1987 for the World Commission on Environment and Development. Sustainable development is the kind of development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs. The main goal is to integrate the three pillars of sustainability that contribute to the achievement of sustainable development.

Environment. Protection and enhancement of natural resources.

Society. The wellbeing of people.

Economy. Sustainable consumption and production.



Most people now recognise that economic activity cannot take place without considering the environmental impacts. For example, the extraction of aggregates for new development has a significant effect on the local environment, which must be taken into account.

The Government's 25-year environmental plan (25YEP) was launched in 2018, and set out an ambition to become the 'first generation to leave the environment in a better state than we found it for our children, and our children's children'. The 25YEP set out a comprehensive delivery plan for the approach to halting and then reversing the decline in nature. Setting out a vision for a 'quarter-of-a-century of action' to help the natural world regain and retain good health, the plan has now started to deliver, and the following is evidence of this delivery in action.

- Wildlife habitats the size of Dorset have been created or restored.
- More than £750 million has been invested in the environment through the Nature for Climate Fund (NCF).
- The UK became the first major economy in the world to commit in law to Net Zero Strategy for the decarbonising of all sectors of the UK economy by 2050 and led international efforts to tackle climate change through our presidency of UN Climate Summit COP26.
- The UK is a world leader in its ambition to work with other nations to address environmental challenges the world is facing, which include the interlinked threat of climate change and biodiversity loss.
- The UK was first to co-design an agricultural programme to help sustainable food production to improve, not just protect the environment.
- A network of marine protected areas across 35,000 square miles of English waters has been established.
- Under the UK's Presidency of the UN Climate Summit COP26, 145 countries (representing over 90 per cent of the world's forests) signed a pledge to halt deforestation and land degeneration by 2030.
- The UK agreed to a new global deal for nature at the UN Nature Summit COP15, which set out a framework for restoring environments.
- The UK has sent a strong global message due to its long-term environmental targets, international leadership on climate and nature, and has evidenced that it takes environmental matters as seriously as Net Zero.

SUSTAINABLE CONSTRUCTION AND THE ENVIRONMENT

The 25YEP has now been refreshed, a commitment which was set into law in the Environment Act 2021 stating it would be reviewed and revised, if needed, every five years to ensure continued progress against the 10 goals of 25YEP. The Environment Act required government to set a suite of legally binding targets for environmental improvement in air quality, biodiversity, water, resource efficiency and waste reduction. These targets are meaningful, ambitious and advised by experts.

To achieve its vision, the 25YEP identified 10 complementary environmental goals, along with specific targets and commitments made in relation to each goal to ensure continual delivery of these targets and the overarching goals. These goals have been used to form the Environmental Improvement Plan 2023 (EIP23), and are laid down as follows:

Goal 1: thriving plants and wildlife

Goal 2: clean air

Goal 3: clean and plentiful water

Goal 4: managing exposure to chemicals and pesticides

Goal 5: maximise our resources, minimise our waste

Goal 6: using resources from nature sustainability

Goal 7: mitigating and adapting to climate change

Goal 8: reduced risk of harm from environmental hazards

Goal 9: enhancing biosecurity

Goal 10: enhanced beauty, heritage, and engagement with the natural environment

The EIP23 addresses how the Government will:

- create and restore at least 500,000 hectares of new wildlife habitats, starting with 70 new wildlife projects including 25 new or expanded National Nature Reserves and 19 further Nature Recovery Projects
- deliver a clean and plentiful supply of water for people and nature into the future, by tackling leaks, publishing a roadmap to boost household water efficiency, and enabling greater sources of supply
- challenge councils to improve air quality more quickly and tackle key hotspots
- transform the management of 70% of our countryside by incentivising farmers to adopt nature-friendly practices
- boost green growth and create new jobs – from foresters and farmers to roles in green finance and research and development.

www **For further information on the ambitious roadmap for a cleaner, greener country and the EIP23, visit the Government website.**

1.2 Important points

The construction industry has a major role in ensuring that the built environment is shaped in a way that delivers a sustainable future and reduces the environmental pressures on our planet from impacts such as climate change, resource depletion and biodiversity loss. For any construction project there will be a number of stakeholders who will have an interest in ensuring that the environmental requirements are being met: investors, clients, neighbours and regulators, to name a few.

Effective environmental management will support your own legal compliance with environmental legislation and should ensure your stakeholder requirements are met. The environmental issues associated with construction are highly regulated. Environment Agencies and Local Authorities across the UK are responsible for ensuring that regulations are being complied with, and impacts that can cause pollution, reduce water supply or create nuisance (such as noise, dust and vibration) are minimised.

1.3 Sustainable construction

There are many reasons for the construction industry to work towards achieving these sustainable development priorities, including:

- The output of the construction industry is enormous; it is worth in excess of £100 billion per year.
- The construction industry accounts for around 6.5% of GDP, employs over 2 million people and supports 280,000 businesses.
- Buildings are responsible for almost 50% of UK carbon emissions, 50% of water consumption, about 30% of landfill waste and 25% of all raw materials used in the UK economy.

In the shift towards a Future Buildings Standard, the UK Government introduced a range of changes to the Building Regulations 2010, including a mandatory 30% cut in carbon emissions for all new homes. These came into force in June 2022, with a one-year transition period to allow for planning applications underway at that time. The changes included amendments to current Approved Documents (ADs) for ventilation (Part F) and conservation of fuel and power (Part L), and new ADs covering overheating (Part O) and infrastructure for charging electric vehicles (Part S). The majority of these changes are aligned to the roadmap and interim measures in the Government's Future Homes Standard and Future Buildings Standard, planned for 2025. A pivotal part is ensuring that all new builds are capable of being net zero in terms of operational carbon.

The Building Regulations will focus on reducing the need to heat and power buildings, and ensure that the UK is reaching its carbon zero goals.

e.g. Government and construction industry response to improving sustainability in construction

- Establishment of the UK Green Building Council (UKGBC) in 2008. Their most recent resources (housing standards playbook and social value in new development) offer guidance for developers and Local Authorities to support sustainable development.
- Improvements that support environmental protection through Parts L and F of the Building Regulations *Conservation of fuel and power* (amended 2022), by the DLUHC Future Buildings Standard, with a new Part O covering overheating in relation to building energy performance, air tightness and water efficiency.
- Establishment of the Waste Resources Action Programme (WRAP), operated between 2000 and 2015; Halving waste to landfill (HWL), and the Built Environment Commitment for clients, contractors, designers and waste contractors.
  Note: *this knowledge base has now been transferred to the Construction Industry Research and Information Association (CIRIA).*
- All new buildings on the central government estate must achieve a Building Research Establishment's Environmental Assessment Method (BREEAM) rating of *excellent (refer to 1.9.1 for further details on BREEAM).*
- Establishment of the Green Construction Board (GCB) as the sustainability workstream for the Construction Leadership Council. The GCB's priority and main focus is to advise on the regulatory, policy and technical framework required to overcome key barriers to the delivery of a zero carbon and zero waste built environment (both buildings and infrastructure), as well as to identify the commercial, jobs and export opportunities that such a clean growth, zero carbon and zero waste economy requires. Following the Infrastructure Carbon Review in 2013, it was identified that infrastructure is responsible for over 50% of the UK's carbon emissions: therefore, PAS 2080 was designed to specifically address the management of carbon in infrastructure. It looks at the whole life cycle of the carbon used on projects, and promotes reduced carbon, reduced cost infrastructure delivery, and a culture of challenge in the infrastructure value chain where innovation can be fostered.
- ISO 20400:2017 (sustainable procurement guidance) provides guidance to organisations on meeting their sustainability responsibilities, by providing an understanding of what sustainable procurement is; what the sustainability impacts and considerations are across various aspects of procurement (such as policy, strategy, organisation and process); and how to successfully implement sustainability within procurement.
- Establishment of the Government Construction Strategy 2016-20, committed to and identifying robust measurements and analysis of sustainability indicators.
- ISO 26000 Social Responsibility provides guidance on how businesses and organisations can operate in a socially responsible, ethical way.
- The Government is committed to the United Nations' sustainable development goals, which have wider reaching objectives beyond construction but also have, at their core, fundamental aims that impact on construction activity.
- The Clean Growth Strategy outlines the aims of the UK Government's vision of a low carbon future that supports reductions in greenhouse gases whilst promoting economic growth through increased efficiencies and innovative solutions.

www

For further information on ISO 20400 visit the ISO website.

For further information on PAS 2080 visit the Carbon Trust website.

1.4 Defining the environment

In its simplest terms, and according to the World Commission on Environment and Development, the ***environment*** is where we all live. More specifically, it can be defined as any physical surroundings consisting of air, water and land, natural resources, flora, fauna, humans and their interrelation.

1.5 Local and global environmental issues

Construction work can affect the environment in a number of ways, most noticeably at a local level where impacts are generally instantly evident. However, it also contributes to wider regional or global issues. The end result of any negative impact on the atmosphere, water or land is usually described as *pollution* or *contamination*.

The surroundings in which a construction project operates includes air, water, land, natural resources, flora, fauna, humans, and their interrelationship, all of which can be impacted by construction activities if not controlled responsibly.

The use of energy, water and natural resources impacts the environment at a local and global level. Construction dust, noise, odours, site lighting and the inconvenience of traffic can each create nuisance, and can all have a significant effect on the neighbourhood and community.

1.5.1 Local environmental issues

The table below provides a summary of typical pollutants that cause local environmental impacts.

Atmosphere	Land	Water
▪ Dust ▪ Exhaust emissions ▪ Gases or vapours ▪ Odours ▪ Noise ▪ Vibration ▪ Light or visual amenity ▪ Smoke ▪ Radiation ▪ Asbestos	▪ Oils and fuels ▪ Chemicals ▪ Lead ▪ Waste and litter ▪ Spillage of materials ▪ Concrete ▪ Asbestos	▪ Silt ▪ Chemicals ▪ Concrete ▪ Contaminated water ▪ Run-off ▪ Effluent ▪ Oils and fuels ▪ Hazardous solid matter ▪ Slurry

! One solution to environmental pollution must not divert the problem to another area (for example, a solution to air pollution should not lead to water contamination).

1.5.2 Global environmental issues

1.5.2.1 Climate change

Certain gases in the atmosphere (principally carbon dioxide, methane, nitrous oxide and chlorofluorocarbons (CFCs)) form an insulating blanket around the planet. This allows the sun's rays through, but prevents some of the heat radiated back from the earth escaping, which warms the planet. This can be likened to the role of glass in a greenhouse, hence the term *greenhouse effect*. Climate change brings various risks for construction, the most obvious of these being higher temperatures and erratic weather patterns, including flooding. It is important to ensure that buildings are designed to adapt to these risks and to mitigate further impacts through energy efficiency and the use of renewable energy.

Over the past century or so, CO_2 levels have risen by 40% since the 1800s, mainly as a result of burning fossil fuels. Although some CO_2 dissolves into the sea,the majority ends up in the atmosphere. Increased levels of CO_2 (and other gases) means that the Earth does not lose all the heat that it absorbs, and therefore the temperature rises. Over the last 100 years, records show that average temperatures have risen by 1°C. As a result of this temperature rise, the air is able to hold more water vapour, now 4% higher than 100 years ago. Increased water vapour has resulted in an increase in the energy stored in the atmosphere, which affects the world's weather patterns.

There have been many indications of climate change occurring from natural disasters, such as:

- the major storm conditions that resulted in severe flooding in New Orleans, USA in 2009
- the hottest average world temperatures were recorded in 2010, and again in 2012, 2014, 2015, 2016, 2020 and 2022
- severe flooding around the world (Australia, India and the UK) in 2011, along with the largest ever number of hurricanes in North America.

What could be the future of climate change? Temperatures could rise by 4% by 2100. Severe weather events in the UK are expected to increase by 30% by the end of the current century. The world's climate is a very complex system, and what might happen is by no means certain. Nothing is certain, but we cannot afford to just wait and see, we have to act now.

1.5.2.2 Acid rain

Acid rain is a collection of loosely related environmental problems involving acid substances and hydrocarbons. Burning fossil fuels, especially coal and oil (mainly in power stations and motor vehicles) produces acid gases (sulphur and nitrogen oxides). The presence of ozone in the lower atmosphere (harmful at this height), formed by the interaction of nitrogen oxides and hydrocarbons, also contributes to acid rain.

1.5.2.3 Deforestation

Deforestation, also known as forest degradation, is the cutting down of trees and forests. With 31% of the world covered in trees, deforestation is a threat to ecosystems around the globe.

The world's forests are being rapidly depleted by logging, slash-and-burn agriculture, development projects (such as businesses, homes, factories, roads, mines and dams), farmland and farm-related industries, and creation of pastures for livestock; forests are also being burned and destroyed by wildfire.

In the 1950s, forests covered about a quarter of the Earth's surface. At the turn of the 21st century, the figure was only one sixth. At our current rate, all rainforests will be gone in 77 years. The Amazon rainforest now emits more carbon dioxide than it absorbs.

According to the World Resources Institute, deforestation increased 12% globally in 2020. Even though the rest of the world pulled back during the global COVID-19 pandemic and the world economy shrunk by around 3.5% in 2020, the destruction of environmentally-sensitive rainforests was on the rise. Without trees, many endangered species become even more vulnerable, and other animals are forced to find new homes. The atmosphere is plagued with higher levels of greenhouse gases that would have otherwise been stored by the trees as a carbon sink. Deforestation is having a detrimental impact on our planet.

Around 200,000 acres of rainforest are cut down or burned each day. That is an area equal to 120 football pitches lost every two minutes of every day, and 28 million acres lost every year.

50,000 species are lost each year as a result of deforestation, while 25% of Western drugs and pharmaceuticals come from rainforest sources, and 25% of all cancer-fighting drugs come from rainforests.

75% of tropical rainforests have lost the ability to properly recover from wildfires and drought, and 15 billion trees are chopped down every year. This equals every person on the planet having 3,000 rolls of toilet paper!

Forests are essential for a healthy world. They play an important role in regulating the global climate by locking up large amounts of carbon dioxide during photosynthesis. Trees prevent erosion, flooding and the formation of deserts.

Forests contain over half the world's plant and animal species and provide a home, fuel and food to many of the world's 150 million indigenous people. They also provide the basis for most medicines and cures discovered to date.

The construction industry uses a large amount of timber and is increasingly aware of the need to use this resource sustainably, hence why certification schemes such as the Forestry Stewardship Council (FSC) and the Programme for the Endorsement of Forest Certification (PEFC) have been established.


The world's forests are being rapidly depleted

1.5.2.4 Biodiversity loss

An ecosystem is made of plants, animals, micro-organisms, soil, rocks, minerals, water sources and local atmosphere interacting with, and over a given period of time sustaining, each other.

The clearing of natural ecosystems, which is usually a result of logging, agricultural or industrial development, or a consequence of war, causes great local disruption to the natural environment.

One major concern about the wholesale destruction of any natural habitat, particularly tropical forest, is that many species may become extinct as the ecosystem disappears.


The loss of natural habitat threatens many invertebrates

On a wider scale, migratory species that use a particular habitat during one season (such as the breeding season) might also become extinct. Loss of biodiversity has a serious impact on the processes of nature. A wide range of biodiversity is an important element for a healthy planet. Animals, plants, trees and micro-organisms offer eco-services that help to purify the water and air, pollinate our crops for food and break down wastes naturally.

The destruction of habitats found in other ecosystems (such as wetlands, oceans and grasslands) has the same consequences as de-forestation. Irresponsible construction can have a devastating impact on biodiversity. In recognition, UK legislators have put planning processes in place to effectively manage this issue.

1.5.2.5 Resource depletion

The world's finite resources (such as coal, natural gas, oil, minerals, soil and metals) are rapidly being depleted. These resources have to be managed in a sustainable way because they are not renewable. Hydrocarbons (oil-based fuels and products) are finite resources because it takes millions of years for them to form: they cannot be replaced naturally within human timescales. Metals are also a finite resource; the amount of any available, as well as chemical and physical usability, is reflected in prices. Forests are a resource that can be regrown, but are currently being depleted *(refer to 1.5.2.3 Deforestation).*


Large scale quarrying of finite resources

Water, especially clean drinking water, is being depleted and even in areas of Europe with regular rainfall this resource becomes limited after a short period of low rainfall. Construction is wholly dependent on the use of resources; without them, development could not happen. The drive for sustainable consumption and production will inevitably increase the costs for scarce resources and lead to technological improvements in the reuse and recycling of existing materials.

1.5.2.6 Ozone layer depletion

The ozone layer is a thin concentration of gas in the upper atmosphere. It protects the planet by filtering out harmful ultra-violet rays from the sun. These can cause skin cancer, eye cataracts, and restrict the growth of plants and other organisms. The layer has been depleted, especially above the North and South Poles, by artificial gases that destroy ozone molecules. The main offender was CFCs, which were used in industry and household goods but have now been banned. Worldwide action on ozone-depleting substances has significantly reduced this issue, with scientists estimating that a full recovery will have been reached by the middle of the 21st century.

1.6 Environmental stakeholders

Environmental management terminology often refers to *stakeholders* who, in simple terms, are interested parties who have either a direct involvement in a construction project or who may be affected by its work. Typical environmental stakeholders for a construction project could include the following.

Stakeholder	Potential interest
Investors	Ensure that the project's sustainable credentials increase value.
Non-governmental organisations (NGOs) (for example, Greenpeace, Friends of the Earth and the Green Building Council)	Publicise environmental practices, both good and bad.
Client or client's representative	Set the project environmental requirements.
Designer	Ensure that the client's environmental aspirations are reflected in the project design.
Planning authority	Approve the design, including relevant environmental requirements and set relevant planning conditions.
Local Authority	Ensure that the works comply with statutory requirements for local air pollution control, including noise and dust, and issue relevant permits.
Environment regulators (for example, Environment Agencies and Local Authorities)	Act as a statutory consultee in the planning process and issue relevant permits and licences for the works (such as discharge consents).
Staff and contractors	Ensure that the project environmental requirements are met and lead by example.
Government funded support organisations (for example, WRAP, the Carbon Trust and charities)	Provide support, advice and good practice and drive environmental improvement.
Residents and community representatives	Participate in the planning and consultation process and ensure that the works bring positive benefits to an area and are carried out without causing nuisance from noise, dust, light and traffic congestion.

Early identification of all relevant stakeholders, together with regular communication and liaison, is important to avoid delays, legal intervention or causing a nuisance to the local community.

Identify all stakeholders and regularly communicate with them on relevant environmental issues.

For further information refer to the community liaison section in Chapter E06 Statutory nuisance.

1.7 Regulatory framework for the environment

Construction sites have a number of legal obligations, many of which relate to protection of the environment. Environmental policy and law is based on guiding principles that define the content of any rules and regulations, and are firmly embedded within legal frameworks.

Prevention is better than cure, so policy measures are introduced to prevent environmental harm rather than remediate environmental problems once they have occurred.

Precaution. Where there is uncertainty over the environmental science of an urgent problem, environmental protection must be prioritised rather than risk an impact occurring.

Polluter pays principle requires that those causing (or potentially responsible for) environmental damage bear the financial costs of any remediation actions. For example, this payment may be made through remediation of contamination, investment in pollution abatement technology and clean production plant, or through environmental taxation and charges.

Legal obligations do not only originate from the UK Government, or its devolved administrations. There are many sources of environmental law, starting at the international level, which are then transposed into UK law. In fact, most environmental law in the UK currently comes from regulations made in Europe.

In the UK there are two general types of law.

Civil law is concerned with the compensation of people who have had harm done to them through no fault of their own.

Criminal law is concerned with the punishment of companies or individuals who have broken the law.

The two relevant sources of law are statute law, which is made by Parliament and implemented in the form of Acts, Regulations and Orders; and common law, which is established by custom and practice by judges through their decisions and the precedents that they set.

In the UK there are three levels of regulation relevant to criminal law.

Statutes or Acts of Parliament are laws that have been agreed by both houses of Parliament in the form of a Bill and have received royal assent. Acts of Parliament are also known as 'primary legislation' because they do not depend on other legislative authority. These Acts are enabling legislation, as they allow for the creation of regulations. The Climate Change Act 2008 is a statute.

Delegated (or secondary) legislation can take different forms.

- Statutory instruments (SIs) add detail to the Acts, through which the Secretary of State can issue specific regulations.
- Orders issuing specific rules relating to a statute or part of a statute (for example, the Carbon Reduction Commitment Energy Efficiency Scheme Order issued under the Climate Change Act).
- Byelaws, which are made by the various tiers of local government, and can cover such matters as the establishment of internal drainage boards.

Guidance. For some pieces of legislation the relevant Government department or regulator will issue guidance (such as those below) on the interpretation and implementation of the regulations.

- **Statutory guidance**, typically in the form of an Approved Code of Practice (ACoP) that sets out how to comply with the law, and gives a clear indication of what is expected.
- **Non-statutory guidance**, typically a circular from a Government department, or technical guidance from either Government or the regulator (for example, Defra *Non-statutory guidance*).

1.7.1 Government policy-making departments

The main Government departments involved in developing and implementing environmental policy and legislation are outlined below. They work with the devolved administrations of Wales, Scotland and Northern Ireland.

Department for Energy Security and Net Zero (DESNZ) is responsible for the mission to replace energy with cheaper, cleaner and renewable domestic sources to enable the UK to become a net zero economy by 2050.

Department for Environment, Food and Rural Affairs (Defra) is responsible for making policy and legislation, and works with others to deliver policies in the natural environment, biodiversity, plants and animals; sustainable development and the green economy; food, farming and fisheries; animal health and welfare; environmental protection and pollution control; and rural communities and issues.

Ministry for Housing, Communities and Local Government (MHCLG) is responsible for implementing planning policy, Building Regulations and building-related environmental standards (energy performance certificates and display energy certificates).

Construction Leadership Council (CLC) has been established to own and oversee industrial strategy for construction. Membership is composed of senior business people representing the main industry bodies and senior representatives of Government departments. The CLC will concentrate on achieving the following joint industry and government commitments from the construction strategy.

- 33% reduction in both the initial cost of construction and the whole-life cost of assets.
- 50% reduction in the time from the outset to completion for new build and refurbished assets.
- 50% reduction in greenhouse gas emissions in the built environment.
- 50% reduction in the trade gap between total exports and total imports for construction products and materials.

Green Construction Board is the sustainability workstream of the Construction Leadership Council, and its priority and main focus is to advise on the regulatory, policy and technical framework required to overcome key barriers to the delivery of a zero carbon and zero waste built environment – as well as to identify the commercial, jobs and export opportunities that such a clean growth, zero carbon and zero waste economy requires. It was established to ensure a sustained, high-level conversation and to develop and implement a long-term strategic framework for the promotion of innovation and sustainable growth. The board owns and monitors the implementation, and will build on the Low Carbon Construction Action Plan and Routemap.

WWW **Further information on these departments can be found on the Government and Northern Ireland Department of Environment (DOE) websites.**

1.7.1.1 Powering Up Britain

This plan is the first to come from DESNZ. Powering Up Britain sets out the department's approach for Britain to be powered by renewables including wind, solar, hydrogen, power with carbon capture, usage and storage (CCUS) and new nuclear plants, and is an introduction to the complementary Energy Security Plan and the Net Zero Growth Plan.

The Energy Security Plan sets out steps to transform the energy system in the UK to make sure the UK is energy independent, secure and resilient and to double Britain's electricity generation capacity by the late 2030s.

The Net Zero Growth Plan sets out the actions to be taken to ensure the UK remains a leader in the transition to net zero by:

- driving investment into industries like offshore wind, CCUS and nuclear
- acting as the annual update against the Net Zero Strategy
- responding to the Climate Change Committee's Annual Progress report
- setting out the package of policies and proposals to meet the carbon budgets up to 2037.

www **Further information on Powering Up Britain can be found on the Government website.**

1.7.2 Environment Agencies

England. The Environment Agency (EA).

Northern Ireland. The Northern Ireland Environment Agency with the Department of Agriculture, Environment and Rural Affairs.

Scotland. The Scottish Environment Protection Agency.

Wales. Natural Resources Wales, linking with the Forestry Commission and Countryside Council for Wales.

The environment agencies exist to provide high-quality environmental protection and improvement. This is achieved by an emphasis on prevention, education and vigorous enforcement wherever necessary.

The aim of protecting and enhancing the whole environment is to contribute to the targets of global sustainable development.

The **enforcement powers** of the environment agencies are set out in their enforcement and sanctions statement and guidance, which outlines the circumstances under which they will normally prosecute. Enforcement powers available to the agencies will vary according to the nature and severity of the environmental offence, but could include the following.

- A formal warning.
- A formal caution.
- Enforcement notices and works notices (where contravention can be prevented or needs to be remedied).
- Prohibition notices (where there is an imminent risk of serious environmental damage).
- Suspension or revocation of environmental permits and licences.
- Variation of permit conditions.
- Injunctions.
- Remedial works (where it carries out remedial works, it will seek to recover the full costs incurred from those responsible).
- Criminal sanctions, including fines, prosecution and imprisonment.
- Civil sanctions, including financial penalties.

The environment agencies can authorise officers to enter premises, including any land, vehicle, vessel or mobile plant. The occupier may need to be notified in advance (other than in the case of an emergency) and entry may only be permitted at reasonable times. Officers must abide by and not go beyond the instructions within the authorisation. In the case of emergencies under Section 108 of the Environment Act persons authorised by the environment agencies have the authority to enter any premises, by force, at any time with or without a warrant, and to:

- take any equipment or materials necessary as evidence
- make any examination or investigation
- direct that the premises are left undisturbed while being examined
- take samples and photographs
- require the production of records to demonstrate and confirm evidence of compliance
- instruct any relevant person to answer questions and to declare the truth of the answers given.

Environment agencies can also carry out formal interviews under caution in accordance with the Police and Criminal Evidence Act 1984. The EA's *Offence response options* document sets out the options available to every offence that the EA regulates.

www **For further information on EA sanctions and offences visit the Government website.**

1.7.3 Local Authorities

Local Authorities are responsible for various environmental pollution control functions, including those listed below.

Air quality. Local Authorities are responsible for the management and assessment of local air quality, including the establishment of air monitoring zones, smoke control areas and the prohibition of dark smoke from chimneys under the Clean Air Act 1993.

Contaminated land. Under the Contaminated Land Regime, Local Authorities have a duty to inspect their land, to formally classify it as contaminated land, and require it to be cleaned up by the owner/occupier. Land can be designated as a **special site** under certain criteria (for example, land seriously affecting drinking water, owned or occupied by the Ministry of Defence or used for certain industrial activity). In these circumstances the land will be regulated by the environment agencies.

Local Authority Air Pollution Control (LAAPC). For less polluting processes requiring an environmental permit (in England, Northern Ireland and Wales) or authorisation (in Scotland), the control of emissions to air alone is exercised by Local Authorities. Local Authorities have a duty to give prior written authorisation for processes under their control. This includes, for example, permits for mobile crushing and screening equipment and concrete batching plants.

Nuisance. Complaints of statutory nuisance (such as noise, dust and odour) are dealt with by Local Authorities. They may serve an abatement notice where they are satisfied that a statutory nuisance exists or is deemed likely to occur or recur *(for further information refer to Chapter E06 Statutory nuisance).*

Planning. Local Authorities have the responsibility of implementing and regulating national planning policy, including environmental impact assessments, tree preservation orders (TPOs) and authorisations for hedge removal. In many cases other regulators will be a statutory consultee, as part of the planning process.

Waste controls. Local Authorities have some powers under waste legislation to stop and search waste carriers and confiscate (and, in some instances, crush) vehicles suspected of waste crime *(for further information refer to Chapter E10 Waste and material management).*

1.7.4 Other regulatory organisations that support the protection of the environment

There are a number of other regulatory bodies (shown below) that have specific regulatory and advisory responsibilities that are relevant to construction sites.

Water companies. The composition and quantity of industrial discharges to sewers are controlled primarily by the regional water companies (frequently referred to as sewerage undertakers) (*for further information refer to discharge consents in Chapter E07 Water management and pollution control).*

Lead Local Flood Authorities. On 6 April 2012, when a further phase of the Flood and Water Management Act 2010 was implemented, responsibility for regulating work on ordinary watercourses in most areas of England and Wales transferred from the EA to Lead Local Flood Authorities (LLFAs). These are unitary authorities where they exist, and county councils elsewhere.

Canal and River Trust. British Waterways ceased to exist in England and Wales and in its place the Canal and River Trust was set up in July 2012 to care for 2,000 miles of historic waterways. In Scotland, British Waterways continues to exist as a public body caring for the canals, operating as Scottish Canals.

Cadw is the historic environment service of the Welsh Assembly Government, having responsibility for designated archaeological and heritage sites in Wales.

Historic England is responsible for protecting historic buildings, landscapes and archaeological sites.

Health and Safety Executive (HSE). There is an overlap between environmental legislation and health and safety legislation regulated by the HSE, including the Control of Substances Hazardous to Health Regulations 2002 and the Control of Major Accident Hazards 2015.

Internal Drainage Boards have powers under the Land Drainage Act 1991 (as amended) to undertake works on any watercourse within their district other than a main river. A board's district is defined on a sealed map prepared by the EA and approved by the relevant ministry. In addition, boards can undertake works on watercourses outside their drainage district in order to benefit the district.

Natural England is responsible for conservation of wildlife and geology, including sites of special scientific interest (SSSIs) and prevention of damage to habitats.

NatureScot is responsible for designated ecological sites, geological and geomorphological sites, and protected species.

Historic Environment Scotland is an executive agency of the Scottish Government and is charged with safeguarding Scotland's historic environment and promoting its understanding and enjoyment.

Department for Communities is responsible for the historic environment in Northern Ireland.

1.8 Construction sustainability assessment tools

Building Research Establishment's Environmental Assessment Method (BREEAM) and BREEAM Infrastructure (formerly the Civil Engineering Environmental Quality Assessment and Award Scheme (CEEQUAL)) are assessment methods used by the construction and civil engineering industry to measure and improve the sustainability of projects.

Clients are increasingly making achievement of these standards part of the project obligations. Many of the assessment standards are directly linked to the objectives and targets set out in the Government's *Strategy for sustainable construction* discussed above.

1.8.1 The Building Research Establishment Environmental Assessment Method (BREEAM)

BREEAM is the world's leading science-based suite of validation and certification systems for sustainable built environment. Since 1990, its third-party certified standards have helped improve asset performance at every stage, from design through construction, to use and refurbishment.

BREEAM is a voluntary green building sustainability rating system established in the UK, and assesses the performances of buildings over a wide range of environmental issues to produce a rating of either Pass, Good, Very Good, Excellent or Outstanding.

A BREEAM assessment uses recognised measures of performance categories and criteria (which cover a range of areas associated with any development, ranging from energy to ecology) set against established benchmarks in order to evaluate a building's specification, design, construction and use. BREEAM assessments are conducted by assessors who score a building or project based on 10 sustainability categories. Each category has a number of credits allocated to it and the overall score is a sum of these.

These 10 distinct BREEAM assessment categories gauge a building's sustainability as comprehensively as possible, and are as follows:

1. Energy: How energy efficient is a building? How much energy usage or wastage is there?
2. Management: How well is a building or project managed? Are there sustainability focused management policies in place?
3. Water: How much water does a construction project require? How much water does the building require to remain operational?
4. Waste: How much waste is produced by a construction project? Where does the waste end up?
5. Pollution: How much pollution results from the building? How can pollution be limited or removed entirely?
6. Health and Wellbeing: What health and safety measures are in place? Is there adequate ventilation and lighting?
7. Materials: Which materials are used to construct a building? Where are they sourced? Are they sustainable?
8. Transport: How accessible is a building to those who live or work there? Is the building easily connected to existing public transport?
9. Land usage: Is this a brownfield or a greenfield site? How is the surrounding environment and wildlife impacted?
10. Innovation: How innovative is the design? How innovative are the building's sustainability policies?

For a building to achieve the highest possible score it must achieve credits in all BREEAM Assessment categories.

The overall score for the assessment will be the percentage of credits achieved in each of the 10 categories, which are then weighted to provide an overall percentage score. The total score will determine the BREEAM rating, as shown on the right.

Assessment	Credits achieved
Outstanding	Minimum 85%
Excellent	Minimum 70%
Very good	Minimum 55%
Good	Minimum 45%
Pass	Minimum 30%
Unclassified	Less than 30%

To ensure that performance against fundamental environmental issues is not overlooked in pursuit of a particular rating, BREEAM sets minimum standards of performance in important areas (such as energy, water and waste).

It is important to bear in mind that these are minimum acceptable levels of performance and, in that respect, they should not necessarily be viewed as levels that are representative of good practice for a BREEAM rating level. To achieve a particular BREEAM rating, the minimum overall percentage score must be achieved and the minimum standards, applicable to that rating level, complied with.

BREEAM third-party certification involves the checking – by impartial experts – of the assessment of a building or project by a qualified and licensed BREEAM assessor to ensure that it meets the quality and performance standards of the scheme. Fees are charged for this service.

www **For further information visit the BRE Group website.**

1.8.2 BREEAM Infrastructure

BREEAM Infrastructure (formerly CEEQUAL) is the recognised industry leader for sustainability assessment tools for infrastructure and civil engineering projects. Its eight categories relate to different areas of sustainability against which the performance of a project is assessed. Projects are awarded a score for each category, and are combined to provide an overall sustainability assessment final rating.

There are six rating levels:

1. Unclassified
2. Pass
3. Good
4. Very good
5. Excellent
6. Outstanding.

The eight BREEAM infrastructure categories

Competent BREEAM Infrastructure assessors (usually a member of the project team who influences and encourages the team to consider sustainability issues at appropriate times) and external verifiers are central to the assessment process. Assessors drive the process and verifiers provide third-party verification of the final rating. The fee scale is dependent on many factors such as project contract value or works, client's or engineer's estimates, and assessment type.

www **For further information visit the BRE Group website.**

1.8.3 Leadership in Energy and Environmental Design

Launched by the US Green Building Council (GBC) in 1998, the Leadership in Energy and Environmental Design (LEED) standard has become widely used both within the US and around the world. In recent years, UK based client groups have begun to ask for LEED certification alongside BREEAM.

Around 2.8 million square feet of buildings around the world are LEED certified every day.

Like BREEAM, LEED is voluntary and it can be applied to any building type and any building life cycle phase. It promotes a whole-building approach to sustainability by recognising performance in important areas of energy and water efficiency, CO_2 emissions, indoor environmental quality and sustainable use of resources.

LEED credits are weighted differently, depending on their potential impact. The greatest weighting is placed on energy and atmosphere, with sustainable sites and materials and resources also receiving a high weighting. A total of 100 base points are available with, additionally, six possible innovation in design points and four regional priority points. There are four levels of achievement: certified (40-49), silver (50-59), gold (60-79) and platinum (80 and over).

There are five different rating systems to cover different types of project, including new construction, LEED for existing buildings, LEED for commercial interiors, LEED for retail, LEED for schools and LEED for core and shell. Most building types are included in one or more of these systems. LEED differs from BREEAM in a number of areas, and contractors should review requirements fully. Some differences are shown below.

- There is no need for an accredited assessor, as the US Green Building Council assesses applications (although an extra credit is available where an assessor is used).
- Design requirements are linked to the American ASHRAE standards whereas BREEAM relates to UK Building Regulations.
- Some credits are calculated using US specific outputs (such as US dollars saved for credits relating to energy).
- Regional priority credits can only be obtained in the US.

In 2018 the Building Research Establishment and the US Green Building Council entered into partnership to raise sustainability standards and deliver greater value.

www **For further information visit the US Green Building Council website.**

1.8.4 RICS SKA rating online assessment tool

In 2005, Skansen initiated a research project with the Royal Institute of Chartered Surveyors (RICS) and AECOM to establish whether it was possible to measure either the environmental impact of fit-outs, or measure or codify good environmental practice on fit-out projects, in order to remove the ambiguity prevalent in the fitting out and refurbishment industry. The result of this was SKA rating, and this assessment tool allows property and construction professionals and SKA assessors to design, specify, rate and certify non-domestic fit-out projects for environmental impact, using the SKA rating fit-out benchmark system. Use of the tool is free and open to all. Projects can be certified by qualified assessors for an additional fee. The SKA assessment process is broken down into three stages.

1. Design/planning.
2. Delivery/construction.
3. Occupancy stage assessment.

www **For further information visit the RICS website.**

1.9 Sustainable use of materials, energy and water – resource efficiency

Materials, water and energy (including transport) are all forms of resource, where efficiency of use has a considerable influence on construction schedules, costs and environmental impact.

Construction projects that use materials efficiently will often have lower construction times and lower costs. Clearly this will lead to greater competitiveness, more repeat business and greater customer satisfaction. It also reduces the amount of resources that are taken from the planet and the amount of waste that the planet receives via landfill from construction work. Inefficient projects can be costly, late, use excessive resources, produce too much waste, and are bad for the corporate image and lead to reduced client satisfaction.

The Government's *Construction 2025* places a top priority on resource-efficient, low carbon construction. To achieve this outcome the project leadership must lead by example and communicate the correct behaviours, at all levels of the project, to emphasise the importance of resource efficiency.

For further guidance on energy management, water management and resource efficiency refer to Chapters E03, E07 and E08 respectively.

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INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.

Scan the QR code to complete a brief survey, and share your feedback on the course.

SITE ENVIRONMENT MANAGEMENT SYSTEMS

Overview

This chapter provides an introduction to environmental management systems, how they can be applied to construction projects and guidance on the environmental documents and records that should be maintained on a construction site.

In many respects these are the systems that capture all of the required detail identified in each chapter of *Construction site health and safety E: Environment* (GE700E).

2.1 Introduction

An environmental management system (EMS) provides an organised, systematic and consistent approach, allowing organisations to address environmental concerns through allocation of resources, assignment of responsibilities, and provision and continuous monitoring and evaluation of procedures, systems and processes.

They help companies reduce their environmental impact, achieve cost savings, comply with legislation and demonstrate their commitment to continual improvement in environmental performance. Being able to demonstrate that their company understands and is managing its environmental impact is an important prerequisite for construction companies when tendering for new business.

It is recognised that adopting standard environmental management policies and practices not only helps in protecting the environment but also brings business benefits in terms of reduction of waste, energy use and improved efficiency.

It also reduces the risk of causing incidents that may result in enforcement action, which could lead to prosecution.

Managing the environment, or environmental impact, is not just for one person in an organisation. It is a process of change for all involved.

An EMS can help all types and sizes of company meet their own environmental and sustainability targets as well as contribute to national targets on climate change, sustainable development, waste, water, emissions, energy, resource efficiency and other environmental issues.

An EMS has a number of important functions.

Help a company understand its environmental impact. It should be a practical tool to identify and describe a company's impact on the environment, manage and reduce these impacts and evaluate and improve performance. An EMS can help with identifying compliance obligations and managing risks and opportunities.

Secure positive environmental outcomes. It should not only document procedures and processes but also focus on improving environmental performance and complying with legal and client requirements.

Improve cost control and improve efficiency. An EMS can also help conformity with customer requirements in the supply chain, enable sustainable procurement policies, enhance a company's reputation, help improve communication with and confidence of employees, regulators, investors and other stakeholders, and improve market positioning, waste minimisation and energy efficiency savings.

To fully contribute to improved environmental performance, a good EMS should include the detail below.

- Be implemented at a senior level and integrated into company plans and policies (top-level leadership and commitment is essential so that senior management understand their role in ensuring the success of an EMS).
- Identify the company's aspects (activities that impact on the environment) and set clear objectives and targets to improve its management of these impacts and the company's overall environmental performance. This also ensures the correct allocation of resources, such as money and staff time.
- Be designed to deliver and manage the organisation's compliance with environmental laws and regulations on an ongoing basis. This will quickly instigate corrective and preventative action in cases of legal non-compliance.
- Deliver good resource management and financial benefits.
- Incorporate performance indicators that demonstrate the effectiveness of the above and can be communicated in a transparent way in annual reports.
- Provide the workforce with ownership of environmental management.
- Enable the identification and correction of any issues, to facilitate continual improvement.
- Be integrated into the entire organisation.

Demonstrating to clients that you have policies and arrangements in place to manage and improve environmental performance will enhance your company's reputation and contribute to obtaining future work.

Whilst larger construction companies will want to demonstrate that they have a robust EMS through certification to a formal standard (such as ISO 14001 *(refer to 2.3)*), smaller companies should consider implementing an EMS as this will help identify their environmental risks and put in place controls to manage them. This will also improve their opportunities in winning future work.

2.2 Important points

An effective EMS helps companies manage their environmental risks, reduce their exposure to potential legal action and improve performance.

This leads to other benefits, such as cost and efficiency savings and improved customer relationships. An EMS can also help companies demonstrate that they have policies, objectives and programmes in place to meet their own sustainability targets, as well as contribute towards wider Government targets.

Important aspects for delivering an effective EMS on site are shown below.

- Identifying the environmental and legal obligations for a project from a review of client documents. This is including, but is not limited to, planning consents and conditions, environmental statements, remediation statements, archaeological statements and ecological management plans.
- Identifying relevant environmental aspects (such as discharges to land, air and water) and meeting these obligations and their associated impacts. This should also include considering emergency situations (such as leaks and spillages).
- Putting the appropriate responsibilities in place to manage and reduce the impact associated with the project environmental obligations and risks.
- Creating an EMS plan that sets out what actions need to be taken, who is responsible for those actions, and identifying the appropriate records that need to be maintained.
- Putting an appropriate inspection, monitoring and audit regime in place to ensure that the project's environmental obligations and their associated risks are being managed effectively.
- Maintaining and retaining appropriate documented information to demonstrate that legal, client and company requirements associated with the project are being met and can be retrieved, if required, for any purpose.

2.3 Types of environmental management system

There are three main recognised standards or schemes.

ISO 14001 is the international standard for EMS, which specifies the components necessary to help organisations systematically identify, evaluate, manage and improve the environmental impacts of their work, products and services. It helps organisations become greener by putting a system of continual improvements in place. It is a framework for establishing performance targets alongside the procedures, systems and reviews that help ensure sustainable business operations.

ISO 14001 requires organisations to meet the following requirements.

- Identify interested parties relevant to the EMS and their needs and expectations.
- Ensure that top management demonstrate leadership and commitment with respect to the EMS.
- Establish an environmental policy relevant to the organisation.
- Identify the environmental aspects associated with the organisation's work and determine the significant environmental impacts.
- Identify applicable compliance obligations associated with the organisation's environmental impact.
- Identify priorities and set appropriate environmental objectives.
- Identify and provide the necessary resources for establishment, implementation, maintenance and continual improvement of the EMS.
- Establish a structure and programme(s) to implement the policy and achieve environmental objectives.
- Implement planning, control, monitoring, preventive and corrective actions, auditing and reviewing procedures to ensure that the policy is complied with and that the EMS remains appropriate, and is capable of adapting to changing circumstances.

EMAS (the EU Eco-management and audit scheme) is a voluntary EU-wide environmental registration scheme that requires organisations to produce a public statement about their performance against targets and objectives, and incorporates the international standard ISO 14001. Refer to the European Commission Environment website for guidance regarding ongoing EMAS registration now that the UK has left the EU.

BS 8555 is a British Standard, updated in 2016, which breaks down the implementation process for ISO 14001 or EMAS into stages, making implementation much easier, especially for smaller companies. The Institute of Environmental Management and Assessment (IEMA) has developed the IEMA Acorn scheme, which enables companies to gain United Kingdom Accreditation Service (UKAS) accredited recognition for achievements at each stage as they work towards ISO 14001 or EMAS. This allows for early recognition of progress against indicators, and can be used effectively to enhance supply chain management by setting agreed levels of performance, certified to a national standard, which have been checked by an independent auditor. Organisations who complete each stage of the scheme are entered on a public Acorn register.

www **For further information and a guide to environmental management systems visit the WRAP website.**

For further details of the ESOS scheme, which is relevant to larger companies, refer to Chapter E03 Energy management.

2.4 Policy, objectives and targets

Requirements for a basic EMS policy are shown below.

- Be appropriate to the organisation, for its size and type of work.
- Be dynamic, flexible and simple so that it is easy to adapt and change, and easy to understand.
- Form part of a balanced overall management system, and be able to interact with other management systems.
- Establish a framework for setting and reviewing environmental objectives.
- Commit to comply with current legislation and other environmental requirements or obligations.
- Commit to pollution prevention and continual improvement.
- Be documented, implemented and maintained.
- Be readily available for employees and other interested parties.

The policy should be endorsed by someone of authority within the organisation, either a director or someone of equal seniority.

Setting targets is vital to a successful EMS and demonstrates the commitment of the business to reduce its impacts and set the path for a programme of continual improvement.

An initial baseline assessment of where the organisation is in terms of its current management of the environment will identify areas for improvement.

The following areas could be considered.

- Measuring and reducing the amount of waste produced.
- Improving recycling rates for different waste streams.
- Monitoring water use, setting reduction targets and implementing and sharing measures.
- Monitoring energy use (electricity and fuel), setting reduction targets and implementing and sharing measures.
- Purchasing services and materials from sustainable sources and/or with recycled content.
- Assisting and improving supply chain knowledge on environmental matters.
- Existing organisational environmental management practices and procedures.
- Industry best practices.
- Legal requirements.

Regular measurement and reporting against company targets will highlight whether they are being met and where future action needs to be focused.

This will allow the company to continually improve its environmental performance and demonstrate to clients, staff and the general public that it is taking its environmental responsibilities seriously.

2.5 Implementing environmental management on site

A company's EMS will define what plans and procedures, together with the appropriate documents, will need to be completed at a site level in what is known as a construction environmental management plan (CEMP). Whatever documents are used to achieve this, there are some clearly defined steps in order to deliver effective environmental management on site. The following five steps should be followed.

2.5.1 Step 1. Identify the project environmental obligations

The first important step in managing project environmental issues is to identify the environmental obligations. An obligation is a requirement to take a course of action, whether legal or moral. In the case of environmental obligations there are a number of potential sources.

- The first of these is legal obligations where individuals and organisations have to follow legal requirements or face prosecution (for example, duty of care under waste law or the protection of wildlife).
- The next concerns contractual obligations and these might relate to performance requirements that a client specifies (for example, BREEAM). Where these are not met a client can resort to a civil action to recover costs for the damages incurred.
- Another type of obligation arises from businesses' corporate and social responsibility and their duty to act in a way that recognises the interests and views of other stakeholders, in relation to the environment. Failure to recognise corporate and social responsibility can result in damage to corporate reputation.

A review of the project's contractual documentation, including any associated planning conditions, also known as Section 106 agreements, should be undertaken to identify the project-specific environmental obligations.

It is essential to have processes in place to identify potential obligations that might arise from different emergency scenarios that could have environmental impacts, and have an effective response plan in place that is regularly tested to mitigate effects if they occur.

2.5.2 Step 2. Identify the project environmental aspects and risks

Following the identification of the project environmental obligations the next step is to identify the associated environmental aspects and risks, together with any relevant opportunities.

Environmental aspects and risks can be present on a specific project wherever there is an interaction with the environment, arising from either raw material inputs or emissions and other outputs to the environment.

The goal of a good environmental management system is to identify what these aspects, risks and opportunities are and to put in place the necessary controls to manage them to within acceptable limits to achieve the desired objective.

- An **aspect** is an element of an organisation's activities, products or services that can interact with the environment. What that means is that you need to look at your activities, products and services that may have an impact on the environment.
- The **impact** is the consequence of not doing something correctly. It is any change to the environment, whether adverse or beneficial, wholly or partially resulting from an organisation's activities, products or services.

e.g. Aspect and impact

Illegally disposing of waste is the aspect, and the associated impact is it being disposed of incorrectly, such as at an unlicensed waste disposal facility.

Discharging water into a drain is the aspect; the impact is discharging water without consent or the discharged water polluting a connected river.

Another common aspect is storing fuel; the impact can be that fuel leaks into the ground and pollutes the ground or an underground drinking water aquifer.

The minimum level of performance is compliance with legal and other requirements. Organisations may also decide that they want to work to good practices where risks are minimised as far as reasonably practicable.

The best way of managing environmental risks is in a systematic way and ISO 14001 *Environmental management systems* provides an internationally recognised framework for doing this.

The production of a project environmental aspects register, identifying the obligations, together with the associated risks and control measures (including emergency situations or possible worst case scenarios), will be an important tool for communicating these issues to contractors and site personnel.

Environmental risks and opportunities should be assessed during the pre-construction phase of a project to ensure that environmental management is properly integrated within the project with respect to these issues.

Environmental risks are identified in the pre-construction phase through a number of sources, including ecology surveys, desk top studies and ground investigations. The client may also be able to provide information on existing environmental issues.

All new surveys and existing information must be included in the pre-construction information and passed to the designers and contractors to enable them to carry out their duties.

2.5.3 Step 3. Identify the environmental responsibilities

Having defined the project environmental obligations and associated risks and opportunities, it will be necessary to identify the main responsibilities for their control.

An environmental management system can involve every person on a project or within an organisation. Certain people within the management system will have specific responsibilities, and it is important that these are clearly defined and set out.

Successful implementation requires involvement from all employees and ownership of individual roles and responsibilities. Examples of functions and positions with specific responsibilities are shown below.

- Directors.
- Environmental co-ordinators.
- Architects and designers.
- Noise specialists.
- Waste co-ordinators.
- Sub-contractors.
- Suppliers.
- Community liaison staff.

It is important to ensure that the necessary lines of communication are defined between those individuals that have prepared the construction environmental management plan and site personnel.

Responsibilities of an environmental manager on a large construction project

- Implementing the requirements of the company EMS.
- Liaising with site teams, the client, stakeholders and a wider environmental team.
- Ensuring that relevant environmental policies are displayed and communicated to all project staff and personnel.
- Ensuring that a project assessment is carried out to identify the main types of work and the associated aspects.
- Ensuring that a construction environmental management plan (CEMP) is developed and maintained to identify the specific environmental requirements and responsibilities and includes legal, client and other relevant issues.
- Ensuring that a waste management plan is developed and maintained to manage waste and includes appropriate responsibilities, waste targets and legal compliance information.
- Ensuring that, as well as waste targets, relevant environmental objectives are set, implemented and monitored in line with the project construction programme or establishment requirements.
- Ensuring that the environmental requirements and objectives are included in inductions, toolbox talks and briefings, and records are maintained.
- Ensuring that specific activity method statements, including those of suppliers, are reviewed to include the relevant project environmental and waste requirements.
- Ensuring that appropriate inspections and monitoring arrangements are put in place to meet the environmental requirements (such as weekly supervisors' inspections).
- Ensuring that appropriate environmental emergency arrangements are identified, implemented and tested, as appropriate.
- Planning and implementing management reviews and audits of the status, adequacy and effectiveness of the project CEMP.
- Managing environmental non-conformances and subsequent corrective actions.
- Reviewing advised changes to environmental legislation and other requirements and taking the appropriate action.
- Ensuring that an environmental management file (EMF) is established to contain appropriate records of the above and is sufficient to meet the environmental requirements.

2.5.4 Step 4. Create a construction environmental management plan

A project construction environmental management plan (CEMP) is a working document that details how a project will mitigate the potential impacts of its construction activities on the environment and the local community.

It is vital for setting out what actions need to be taken and who is responsible for them. It will contain information including method statements, legislation, performance requirements and environmental risks. The CEMP is usually compiled by an environmental specialist, and the typical contents of a project CEMP will reflect the organisation's internal policies and environmental culture, as well as providing a practical set of guidelines for project processes during construction.

It may include, but is not limited to, the following:

- Environmental policies and EMS requirements.
- Project environmental objectives and targets.
- Environmental roles and responsibilities.
- Environmental risk assessments.
- Development of method statements.
- Site waste management plan.
- Environmental emergency response/action plans.
- Consultation.
- Environmental inductions, training and awareness.
- Environmental monitoring, measuring, inspections and audits.
- Environmental incident and investigation reports.
- Project environmental records.
- Compliance with current legislation.

2.5.5 Step 5. Monitor and inspect

After implementing a project CEMP a robust monitoring and inspection regime should be put in place to ensure that the identified risks and opportunities are being managed and that legal and contractual requirements are being met. Examples of monitoring and inspection work for risk management and compliance include the following.

- Noise level testing.
- Water sampling and analysis.
- Air monitoring.
- Dust monitoring.
- Duty of care checks.
- Soil testing.
- Microbial monitoring.

Monitoring and inspection can also take place to assess performance delivery and the achievement of performance objectives that have been set by a project. This may include monitoring and inspection of energy, carbon, water, resources and material use efficiency.

Monitoring and inspection work can include the following.

- Director tours.
- Site management tours.
- Supervisor tours.
- Internal audits.
- External audits.
- Supplier audits.
- Water monitoring.
- Energy usage.
- Resource use measurement.

2.5.6 Environmental performance improvement

Environmental performance improvement is vital to successful construction projects. Efficient use of energy, carbon, water, raw materials and waste reduction will bring multiple benefits to projects and contribute to cost savings. To achieve performance improvement requires a well-planned and structured management system, the right leadership and a mindset across the project that results in behaviours that are focused towards resource efficiency.

Regular project reviews involving all site personnel that are identified in the project CEMP, including sub-contractors and suppliers, will provide the opportunity to discuss environmental issues and how to improve performance. These reviews need not be separate, formal meetings and could be an agenda item of standard management meetings.

2.6 Environmental documentation

The project environmental requirements will dictate what environmental documentation is required to demonstrate compliance.

It is likely that a company will have a standard project recording system and it would be good practice to ensure a section of this is reserved for environmental records.

Some likely documents that will need to be retained on site

- Initial review and identification of project environmental obligations.
- CEMP, including schedule of aspects, risks and details of their control.
- Waste management plan.
- Details of emergency environmental arrangements, including drainage plans and location of inspection chambers.
- Copies of environmental licences and consents, which could include, for example, the following.
  - Waste carrier licences.
  - Environmental permits for waste transfer stations.
  - Hazardous waste site registration.
  - Environmental permits for the operation of mobile crushing equipment.
  - Environmental permits for the operation of mobile soil treatment equipment.
  - Environmental permit exemptions for the reuse of construction waste.
  - Environmental permits for discharge.
- Waste transfer documentation.
- Environmental inductions, briefings and toolbox talks.
- Environmental consultations, including meetings and formal correspondence with regulators.
- Environmental incident reports and non-conformance reports and breaches.
- Environmental work instructions and operational procedures.
- Environmental monitoring and inspection records, which could include, for example, the following.
  - Weekly site inspection records.
  - Audit reports.
  - Water sampling records.
  - Soil and waste sampling records.
  - Dust and air quality monitoring, including visual inspections.
  - Visual records of sensitive area protection, including fencing.
  - Records associated with the achievement of project environmental objectives and key performance indicators.

A number of these documents will need to be retained for legal purposes, such as waste transfer documentation, which must be retained and accessible for:

- two years for non-hazardous waste
- two years for season tickets (tickets that can be set up for multiple transfers of the same type of waste over a 12 month period)
- three years for hazardous waste transfer notes, and hazardous waste consignment notes (different retention periods apply for consignees (receivers) of hazardous waste; refer to further detail in the hazardous waste guidance)
- six years if you are a landfill operator for non-hazardous waste (for landfill tax purposes)
- the lifetime of your permit if you are a landfill operator for hazardous waste
- the lifetime of an environmental permit (when the permit is surrendered, the regulator often requires a history of the types of waste received).

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Supporting INFORMATION

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Summary of energy management legislation and guidance

This list is not exhaustive and only includes legislation mentioned in this section of GE700.

Legislation and guidance	Enforcement agencies*				
	EA	LA	NIEA	NRW	SEPA
Acts (primary legislation)					
Climate Change Act 2008	✓		✓	✓	
Climate Change (Scotland) Act 2009					✓
Environmental Protection Act 1990	✓		✓	✓	✓
Regulations (secondary legislation)					
Building Regulations (Parts F, L and O) 2010		✓			
Building (Scotland) Regulations 2004					✓
Energy Performance of Buildings (England and Wales) Regulations 2012	✓		✓	✓	✓
Energy Savings Opportunity Scheme Regulations 2014	✓		✓	✓	✓
EU Energy Efficiency Directive					
Streamlined Energy and Carbon Reporting Framework	✓		✓	✓	✓
Guidance					
Building Research Establishment publications and guidance					
Building Standards Technical Handbook		✓			
Net Regs Environmental guidance					
Information and guidance on websites	✓	✓	✓	✓	✓

***Key**

EA	Environment Agency
LA	Local Authorities
NIEA	Northern Ireland Environment Agency
NRW	Natural Resources Wales
SEPA	Scottish Environment Protection Agency

Overview

Global warming and climate change have come to the fore as sustainability issues. Current energy production through the use of fossil fuels is a main contributor to UK carbon dioxide (CO_2) emissions. Reducing the reliance on fossil fuels, and the subsequent production of CO_2, is a growing concern for governments and many other organisations around the world. The construction industry as a whole needs to adapt its design and construction processes to meet clients' expectations and to prevent further environmental impacts, now and in the future.

This chapter gives a general overview of the UK's legally binding energy and carbon reduction targets. It provides guidance on actions that help construction companies to measure and reduce energy consumption and the associated carbon footprint during both the construction process and the occupation of the buildings and infrastructure on completion.

3.1 Introduction

Climate change is now seen as the defining challenge of this era. It is recognised that human activity (through the burning of fossil fuels in energy production, manufacturing and use in buildings and transport) and the generating of waste create greenhouse gases that contribute to global warming and climate change. Buildings, and the construction activity to produce them, contribute a significant amount to the overall carbon footprint from emissions. Carbon dioxide accounts for about 76% of all emissions from the seven main greenhouse gases. Almost half of these emissions are associated with buildings. The construction industry has a major role to play in improving energy efficiency and reducing greenhouse gases and costs.

Action to address the problem of global warming started in 1988, when the intergovernmental panel on climate change (IPCC) was established. The United Nations Earth Summit at Rio de Janeiro in 1992 was, however, the turning point in the fight against climate change. In the intervening years policies have evolved between developed countries around the globe, agreeing to make legally binding cuts in seven greenhouse gases, including carbon dioxide.

The UK has implemented the Climate Change Act 2008, which commits the UK government by law to reduce greenhouse gas emissions by at least 100% of 1990 levels (net zero) by 2050. This includes reducing emissions from the devolved administrations (Scotland, Wales and Northern Ireland) which currently account for about 20% of the UK's emissions.

In addition, in 2009 the UK set a target for 15% of all energy to be generated from renewable sources by 2020 from agreement at a European level. Recent Government figures indicate that these targets will be exceeded, with 2017 figures for reduction in greenhouse gases being reported at 43% (9% above the 2020 target) and renewable sources of energy being exceeded by almost 100% with figures of 28.1% being reported in 2018.

In 2019 greenhouse gas emissions from electricity generation were down 12% on 2018 levels and 71% lower than 1990 levels, the share of low-carbon electricity generation has risen to 59.3% in 2020 with renewables at 43.1%. In 2021 the UK Government introduced the Net Zero Strategy, which set further targets including a commitment to fully decarbonise our power system by 2035.

The UK Government's *Construction 2025: industrial strategy for construction*, published in July 2013, and the more recent Clean Growth Strategy set out an aspiration for a reduction in greenhouse gas emissions in the built environment of 50% by 2025, and 80% by 2050. Alongside this, the Green Construction Board has developed the Low Carbon Routemap enabling customers to understand the policies, actions and key decision points required to achieve the UK's 80% reduction target, to develop market and technology-based plans to secure the jobs and growth opportunities from driving carbon out of the built environment, led by the Green Construction Board.

www **For further details about Construction 2025 and the Clean Growth Strategy, visit the Government website.**

3.2 Important points

Energy is a valuable resource. Reducing the amount used and developing more efficient methods of generating energy can provide significant savings and reduce the emission of greenhouse gases, which contribute to climate change. A construction project can save energy in two main ways. First, by managing the energy used for the construction phase and second, by constructing the project in a way that ensures the completed structure achieves its optimum energy performance for its lifetime use *(for more information refer to 3.6.1.2 Quality and workmanship later in this chapter)*. The following site practices should be employed to reduce energy consumption and greenhouse gas emissions.

- Engage with distribution network operators and energy suppliers as early as possible to obtain a connection to the site to avoid the use of petrol or diesel generators.
- Ensure that all plant and equipment is switched off when not required. Reinforce through monitoring and inspections.
- Ensure that all plant and equipment is regularly maintained.
- Put an efficient transport policy and logistics plan in place, to reduce the need for lorry movements and double handling.

- Consider energy efficiency technologies for site accommodation (for example, LED lighting, passive infrared (PIR) sensors and double glazing).
- Monitor and regularly report site energy and fuel use, and positively reinforce reductions through awareness and toolbox talk sessions..
- Use more sustainable materials. Ensure eco-friendly building materials (such as cork or sheep's wool insulation) are identified at the planning and design stages.
- Eliminate wastage by reusing building materials and components.
- Replace high-emission materials with sustainable timber.
- Use low-carbon concrete instead of traditional cement.
- Switch to electric, hydrogen and hybrid-powered plant and machinery from those powered by fossil fuels.

Construction detailing and quality of workmanship on site can have a major impact on the performance of the building in use. Gaps in insulation at junctions between windows and doors can cause cold bridging, reduced air tightness, increased heat loss and subsequent energy use. Building performance can also be supported and optimised through effective handover and commissioning processes called *soft landings*. Follow up and seasonal commissioning, including the appropriate training of operational staff in the use of the building's control systems, can also significantly improve the performance of a building in use.

3.3 Government incentive and certification schemes

Many mandatory and voluntary initiatives have been introduced by the UK and local governments, as well as the European Union, in response to the risk posed by climate change.

e.g. **Examples include the climate change levy on fossil fuels, revisions to Part L of the Building Regulations 2010, the Energy Performance of Buildings Directive, the EU Emissions Trading Scheme (ETS), the Renewable Heat Incentive, the Carbon Reduction Commitment (CRC) Energy Efficiency Scheme, Energy Savings Opportunity Scheme (ESOS) and the Low Carbon Construction report.**

3.3.1 Streamlined energy and carbon reporting framework

The *Streamlined energy and carbon reporting* (SECR) framework was introduced by the Department for Business, Energy and Industrial Strategy in April 2019, making mandatory energy and carbon reporting easier and aims to incentivise energy efficiency and cut emissions in large energy users in the UK's public and private sectors.

The framework applies to circa 11,000 companies across England, Northern Ireland, Scotland and Wales. It requires companies to measure and report their carbon emissions and to purchase carbon allowances in line with their carbon footprint. It applies to all quoted companies (those listed on the public exchange), and to larger UK incorporated unquoted companies with at least 250 employees, or an annual turnover greater than £36m and an annual balance sheet greater than £18m.

However, organisations that use a low level of energy (40MWh over the reporting period), public sector organisations and those undertaking public activities such as charities, universities, hospitals and academies are not required to submit SECR reporting. The government is encouraging voluntary participation to support transparent ESG reporting for these currently exempt organisations.

www **For comprehensive guidance on the *Streamlined energy and carbon reporting* framework visit the Government website.**

3.3.2 Energy Savings Opportunity Scheme Regulations 2014 (ESOS)

ESOS is a mandatory energy assessment scheme for organisations in the UK that meet the qualification criteria of being a large undertaking, in line with one or both of the following conditions.

- Employing 250 or more people.
- Having an annual turnover in excess of 50 million euros (£44,845,000) and an annual balance sheet total in excess of 43 million euros (£38,566,700).

The Environment Agency is the UK scheme administrator. ESOS applies to large UK undertakings and their corporate groups. It mainly affects businesses but can also apply to not-for-profit bodies and any other non-public sector undertakings large enough to meet the criteria.

Organisations that qualify for ESOS must carry out assessments every four years. These are audits of the energy used by their buildings, industrial processes and transport to identify cost-effective energy saving measures.

Qualifying organisations must notify the Environment Agency by a set deadline that they have complied with ESOS obligations. Various penalties apply for a breach of these regulations. In particular, in relation to providing a false or misleading statement, and to a failure to:

- notify
- maintain records
- carry out an energy audit
- comply with a notice.

Organisations caught under ESOS can now demonstrate compliance by implementing ISO 50001. If this route is chosen, early action is advised as it often takes well over a year for companies to achieve certification.

For further details of ESOS and qualification criteria refer to the Department for Business, Energy and Industrial Strategy and the Environment Agency.

3.3.3 PAS 2080 Carbon management in infrastructure

PAS 2080 *Carbon management in infrastructure* is a standard for managing whole-life carbon in infrastructure and introduces a joined up approach to the way industry evaluates and manages whole-life carbon emissions to reduce carbon and reduce costs. It aims to overcome disconnects in the value chain, embed effective whole-life carbon management approaches and encourage the right behaviours and approaches from clients, contractors, designers and product suppliers to deliver reduced carbon and reduced cost infrastructure.

3.4 Managing energy on site

Conserving energy during any construction activity is an ideal way to cut costs and save money. The main areas for action include on-site construction (energy use, plant and equipment) and site accommodation, transport associated with the delivery of materials and removal of waste, business travel and corporate offices. There are a number of measures that can be undertaken by companies to help reduce their energy use and carbon emissions.

- Ensuring sites connect to the electricity supply as early as possible to prevent lots of equipment running on fuel-driven generators.
- Using hybrid generators that switch off diesel generators when the power load is low to reduce fuel consumption and pull energy from internal storage batteries.
- Using sustainable or recycled products for site set up, logistics and enabling works.
- Using products with a lower carbon footprint.
- Installing eco-friendly site accommodation (for example: double glazing; door closers; waterless urinals; energy-efficient lighting (such as LEDs) and heating (controlled wirelessly); daylight controls; mechanical heat recovery ventilation; solar thermal heating systems; solar photovoltaics; passive ventilation; air-source and ground-source heat pumps; and green roofs).
- Efficient use of construction plant and equipment through induction and training (such as turning off when not required and keeping plant well maintained).
- Good practice energy management on site (for example, festoon low-energy light bulbs and turning task lighting off when not in use).
- Using off-site consolidation areas to facilitate smaller loads being combined and loaded onto larger vehicles (reducing the number of vehicle movements to and from site).
- Sharing transport to reduce the number of vehicle movements to and from site. Vehicle sharing can be planned by site staff, and project suppliers may permit the use of their parking areas as car-pooling points.
- Switch to electric, hydrogen and hybrid-powered plant and machinery rather than fossil fuels.
- Provision of electric charge points for encouraging the use of electric vehicles.
- Fuel-efficient driving through driver training.
- Good practice energy management in company offices.

Meter for monitoring electricity use and cost

Standards have been introduced to manage environmental issues, and ISO 50001 is specific to energy management. It provides a practical way to improve energy use, through the development of an energy management system, to meet the following.

- Develop a policy for more efficient use of energy.
- Fix targets and objectives to meet the policy.
- Use data to better understand and make decisions concerning energy use and consumption.
- Measure the results.
- Review the effectiveness of the policy.
- Continually improve energy management.

ISO 50001 certification can be used to comply with ESOS, as this is sufficient to constitute an ESOS assessment *(refer to 3.3.2)*.

3.5 Measuring energy and carbon

Measuring overall energy usage and being able to identify where energy is being used, and in what form, is a prerequisite to starting an energy and carbon management programme. To do so will require the use of meters to measure energy usage, and on large sites this may involve sub-meters. Sites may be required to measure journeys to and from the project associated with deliveries and business travel and measure the embodied energy associated for the materials they are using.

Embodied energy is an accounting methodology that aims to find the sum total of the energy necessary for an entire product life cycle. This life cycle includes raw material extraction, transport, manufacture, assembly, installation, disassembly, deconstruction and/or decomposition.

To measure the carbon footprint associated with energy use will require the use of conversion factors for each energy type. Different methodologies produce different understandings of the scale and scope of application and the type of energy embodied. Some methodologies account for the energy embodied in terms of the carbon that supports economic processes. Three levels of assessment are currently being used.

Scope 1 are the emissions from sources under the immediate control of the company.

Scope 2 are the off-site emissions from the purchase of electricity.

Scope 3 are the off-site emissions from the company's supply chain or from products sold by the company.

The Greenhouse Gas Protocol developed by the World Business Council for Sustainable Development (WBCSD) and the World Resources Institute (WRI) is the global standard for the measurement and reporting of Scope 1, 2 and 3 carbon emissions.

The Environment Agency has a carbon calculator for construction work, which is hosted on the Government website. This calculates the final carbon impact of different material and transport options, and the footprint of an overall project. It also considers personal travel, site energy use and waste management. The tool has a number of benefits, some of which are shown below.

- Helping to assess and compare the sustainability of different designs, in carbon dioxide (CO_2) terms, and influencing design selection at the options appraisal stage.
- Helping to highlight where big carbon savings on specific construction projects can be made.
- Calculating the total carbon footprint from construction and helping to reduce it.

The tool was developed by the Environment Agency for its own construction work (predominantly river and coastal construction projects). However, other construction clients, contractors and consultants may find it useful when assessing their own work.

For carbon reduction case studies refer to Appendix A.

Measuring and reporting of site energy use is also a requirement under the management credits of BREEAM. The BREEAM Industrial assessment scheme requires both energy and carbon assessments.

The European Network of Construction Companies for Research and Development (ENCORD) has established a *Carbon measurement protocol* to assist in the reduction of emissions from current and future construction work and operations. The document identifies the intended users of the protocol, the sources of emissions over which a construction company may have influence, and the method of measuring these emissions. Guidance is also provided on reporting methods at a company and project level, with a view that companies will report their emissions publicly.

As the energy supply becomes greener, the proportion of energy locked up in building materials increases. The greatest carbon use in the life cycle of a building is during use. Therefore, designers need to think about both embodied and operational energy of the building.

Cleaner construction machinery for London: a low emission zone for non-road mobile machinery (NRMM)

For projects in London an inventory of all NRMM should be kept on-site, stating the emission limits for all equipment. All machinery should be regularly serviced and service logs kept on site for inspection. This documentation should be made available to local authority officers as required.

The NRMM register is an online inventory and details of all NRMM with a net power between 37 kW and 560 kW should be recorded, along with an indication of the proposed duration of use, no matter how short or long this may be, when the machinery is delivered to the site. NRMM is currently only active in Greater London but there are plans to expand it to other cities. From 1st January 2040, only zero-emission machinery will be allowed to operate within this zone.

The NRMM is part of London's Supplementary Planning guidance, detailed in *The control of dust and emissions from construction and demolition.*

www **For further information visit the NRMM webpage.**

3.6 Delivering energy performance in buildings

When considering energy use and emission reduction, this chapter has so far focused on construction site activities. Equally, there are important contributions that can be made to ensure completed buildings are energy efficient: warm in winter and cool in summer, with excellent indoor air quality, low energy bills and low carbon emissions. Every building will consume energy and may create emissions throughout its operational life. It is therefore important that buildings and projects are completed to ensure optimum energy performance and that clients help to reduce the costs and carbon emissions associated with this energy use.

3.6.1 Improving performance

3.6.1.1 Low carbon renewable energy

Transitioning from a high to low carbon construction project can be achieved by strategies focusing on:

- good quality of workmanship on site
- promotion of good practice
- waste reduction
- use of low-embodied energy, local products and materials
- use of recycled and recyclable products and materials
- designing for deconstruction
- on-site energy generation and storage
- reducing the use of vehicles and switching to electric
- high levels of insulation
- low infiltration rates
- low water and power-consuming appliances
- carbon capture and storage
- passive design techniques
- using alternatives to on-site diesel, such as bio-diesel.

Photovoltaic power

Biomass heating

Ground source heat pump

The following types of low carbon renewable energy are increasingly being used on construction projects.

Renewable sources of electricity

- Solar photovoltaic (PV).
- Wind.
- Hydro-electricity, including tidal energy.
- Electricity generated from anaerobic digestion (AD).
- Electricity generated from combined heat and power (CHP).

Renewable sources of heat

- Biomass heating.
- Solar thermal hot water.
- Ground-source heat pumps.
- Air-source heat pumps.
- Heating generated from biomass-fuelled CHP.

In addition to renewable energy, a concept known as *district heating* or *heat networks* is increasingly being utilised. It works on the principle of supplying a number of homes and/or public buildings from one centralised location, with the heat being transferred through highly insulated pipes to each building.

Renewable energy reduces the overall carbon emissions of the building or project. However, the correct way to use the technology is to first consider energy reduction approaches (such as more insulation, better performing windows and glazing, or more efficient building services and domestic appliances).

Renewable energy can then be used to further reduce emissions and costs from the remaining energy needed. Renewable energy will, however, be a vital element of achieving zero or near zero carbon energy requirements set out in improved standards, such as the Building Regulations and BREEAM.

3.6.1.2 Quality and workmanship

As building regulations and air tightness standards for new and refurbished buildings become more demanding, the quality of materials and the workmanship, skills, knowledge, training and experience of the workforce required to install them have a significant impact on energy efficiency. It is therefore important that all site personnel understand how energy-efficient construction can be achieved, to deliver good energy performance in reality.

Incorrect detailing and substituting materials for inferior alternatives (which may look the same but have lower thermal performance) result in buildings that will not meet the expectations of design and thus perform poorly throughout their lifespan, resulting in a significant increase in energy use and running costs.

Insulation quality. Poorly-fitted insulation causes cold bridging (an area where there is a gap in the insulation which is colder than the main areas in a building) and heat loss (the term used to describe the measure of heat escaping from the inside to the outside of a building). This can have a significant negative impact on a building's energy performance. Poorly-fitted cavity wall insulation board has been measured as creating a 300% greater heat loss than designed values.

Air tightness. Air gaps, caused by poor detailing, unsealed service penetrations and poorly fitted doors and windows, allow more air infiltration than the designed values. These gaps will increase heat loss or heat gain, leading to a higher demand for additional energy for heating or cooling.

The lower the air tightness testing result in $m^3/(hr.m^2)$, the lower the unintended heat loss or gain will be (for example, an air tightness testing result of 1 $m^3/(hr.m^2)$ equates to a lower unintended heat loss or gain than a result of 10 $m^3/(hr.m^2)$ would).Performance greater than 10 times the Building Regulations' minimum standard has been achieved on many UK construction projects.

Cold bridging. Significant heat loss can be caused through cold bridges, produced by poor design, changes in the design details or poor workmanship. This heat loss can create condensation problems and damp when the building is in use. Installing insulation to provide a continuous barrier, particularly at junctions of walls, roofs, floors and around windows, is important in avoiding this.

An example of poorly-fitted cavity wall insulation board (Image supplied by Willmott Dixon)

Testing for air tightness (Image supplied by Willmott Dixon)

Thermal image showing excessive heat loss above second floor window heads, indicative of poor continuity of thermal insulation (Image supplied by Willmott Dixon)

3.6.2 Achieving energy efficiency

Buildings should work as an energy system in which heat gains and losses are always in balance. The higher the loss or gain from the building fabric, the more energy is needed to heat or cool it. Buildings lose heat through the external envelope in two ways: firstly, by a complex mix of conduction, convection and radiation through the materials and air spaces in the construction, and secondly via direct air leakage from inside to outside. This means that the design of the building envelope is critical.

Insulation layers and the detailing at junctions need to be designed and constructed with considerable care, based on a good understanding of the heat loss mechanisms involved. In addition, heating or cooling equipment needs to be correctly sized, energy efficient and the whole service system designed, installed, commissioned and operated to achieve maximum overall efficiency.

Passivhaus

Passivhaus is an international design standard which delivers high standards of comfort and health, and minimises the energy use of buildings. Passivhaus buildings use very little energy whilst providing excellent air quality and high levels of comfort for the occupants. Attention to detail in design and construction are guided by principles developed by the Passivhaus Institute in Germany. Buildings designed and built using these exacting standards can be certified through a quality assurance process.

www **For further information visit the Passivhaus Trust website.**

3.6.3 Closing the performance gap

There is substantial evidence to suggest that there is a huge gap between how much energy a building is designed or predicted to use and its actual energy use, which is often much higher – in some cases it can be five to 10 times higher than compliance energy consumption calculations conducted during the design stage. The difference between anticipated and actual performance is referred to as the performance gap.

Although some of this can be attributable to variations in use, which is difficult to predict, a large element is related to the technical under-performance of construction materials and services due to design and construction failings; this is referred to as the *performance gap*. One of the important contributors to the performance gap is a lack of understanding of energy efficiency within the built environment workforce.

It is now common for building performance to be either measured during construction (for example, air tightness tests) or after the building is complete, to identify how it performs whilst in use, rather than just relying on a statement of the design intent.

www **For further information and a copy of the *Sustainable building training guide* visit the Construction Leadership Council's website.**

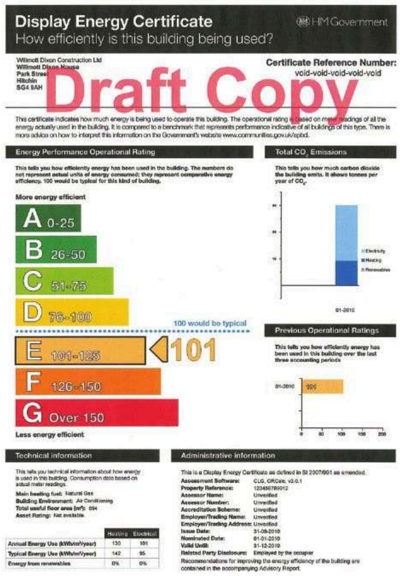
3.6.4 Understanding performance

Measuring a building's performance is achieved by using the Building Regulations Part L, or Building Standards in Scotland, and carrying out a standard assessment procedure (SAP) calculation. In Scotland, a heat loss calculation may also be utilised. The SAP calculation generates the information that informs at what grade the energy performance certificate (EPC) will be issued. EPCs are issued using a grading system from A to G to reflect the building's energy performance in operation, with A being the most efficient. Certification processes (such as BREEAM and LEED) are also useful in performance measurement.

All of these systems indicate how the energy efficiency of the design and construction of the project should perform **on paper**. Understanding how buildings are actually performing during occupation or operation is the first step in the improvement process.

An important factor is the regulations' methodology for checking how sustainable designs and performance promises work out **in reality**. There are many ways that this can be measured for buildings, but one of the most common in the UK for public buildings is the display energy certificate (DEC). The DEC evaluates how much energy the building actually uses during 12 months of occupation, and compares this performance with the typical use for buildings of a similar type.

The way that projects are actually performing is not always known to the contractors building them, creating a risk of them becoming disconnected. Research has shown that actual performance differs from clients' expectations of the regulation levels by a significant degree. This is known as the ***performance gap***.



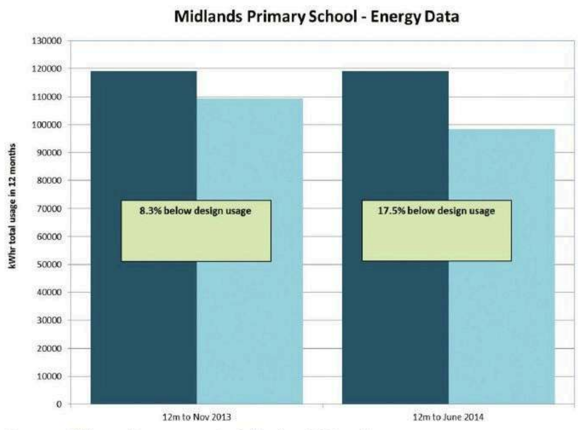
Display energy certificate
(Image supplied by Willmott Dixon)

Industry bodies (such as the UK Green Building Council and Zero Carbon Hub) have published numerous case studies explaining these issues in detail. The important point for the industry is to know how it can act to improve the energy performance of a building at design, construction, commissioning and handover stages of the project. The Building Performance Network (BPN) is an initiative that was established to improve a building's performance whilst in operation, to address in-use performance. The BPN brings together individuals and organisations who have an interest in improving building performance in operation.

3.6.5 Commissioning, handover and aftercare

Making sure that a building and its mechanical and electrical services are working properly and are handed over to the client in an effective way are vital steps to ensure it operates as it was intended to, and that energy consumption in use is minimised. The process, known as *soft landings*, is a step-by-step approach to ensuring that this is achieved as part of the project, includes a building performance evaluation, and is integral to targeting net zero carbon.

This approach is based on structured collaboration between designers, clients, future occupants and operators of a building to ensure that buildings being handed over are comfortable, easy to use and have systems in place to minimise energy use and maintenance costs. The soft landings process will contribute towards achieving net zero carbon buildings by helping to close the performance gap.



Energy efficiency improvements following POE actions
(Image supplied by Willmott Dixon)

The soft landings' six-phase approach is as follows.

Phase 1: Inception and briefing. This establishes client requirements and success criteria, commits all participants to complete the project after handover and allocates responsibilities.

Phase 2: Design. Brings the project team together to review previous comparable projects, detail how the building will work for both the manager and the individual users. Agreeing the energy, the metering and monitoring strategies and the approach to commissioning.

Phase 3: Construction. Ensure that the project team is fully aware of the project's success criteria. Ensure that the facilities manager and the end user are closely involved in the overall project and decisions which affect operation and management of the delivered building

Phase 4: Pre-handover. Scaled handover enables operators to spend time understanding interfaces and systems before occupation. Ensure the building management system is set up the way the client intended including energy data reconciliation, data storage and the energy monitoring software. The metering must be checked to ensure it's working properly and will deliver real world insights into the energy use.

Phase 5: Initial aftercare. Project team should be on site for a period of time to spot emerging problems and issues, and to check that they meet the occupants' expectations and actual requirements.

Phase 6: Years 1-3 extended aftercare and POE. This is a period of longer term, less intrusive monitoring and support. Ensure that the energy monitoring is set up and working well and conduct systematic post-occupancy evaluation (POE) after 12 months, with a final project review at month 36.

For further details of the soft landings approach and free guidance visit the BSRIA website.

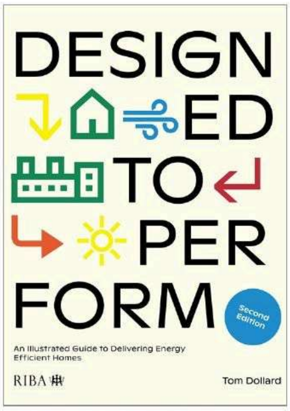
UK construction is already delivering quality projects every day, on time and on budget. Focusing on these issues will make 'on performance' the next major achievement.

3.6.6 Designed to Perform: An Illustrated Guide to Delivering Energy Efficient Homes

An illustrated and informative practical design guide to delivering better energy performance in all types of new-build homes, *Designed to Perform: An Illustrated Guide to Delivering Energy Efficient Homes* addresses construction quality and performance of new homes.

The performance gap between predicted and actual energy use in new homes has been identified as a key problem by the government and industry experts. This guide introduces the concept of the performance gap and highlights clear issues and solutions to help architects improve their detailing at design stage.

The guide has annotated details with photos taken from live construction sites.



The Designed to Perform book

Appendix A – Carbon reduction case studies

To access the *Construction carbon calculator*, referred to in some of the case studies below, visit the Government website.

Sandford bridge

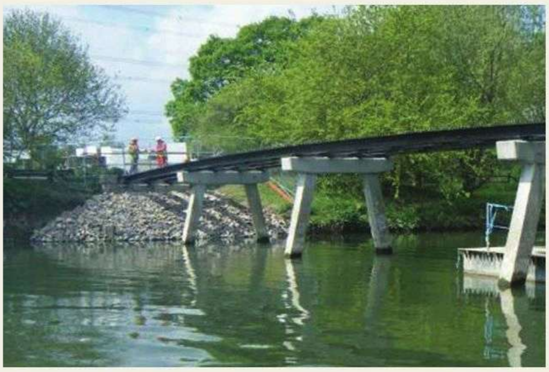
Background. The Sandford bridge project (£380,000) spans the Sandford lock bypass channel on the River Thames. The bridge is primarily used for access to the lock house and the lock structure, but additionally there is a 3 m wide roadway spanning 40 m across the river. The aim of the bridge refurbishment was to increase the carrying capacity from 3 to 18 tonnes.

Reducing the carbon footprint. The design team were interested in minimising the environmental impact and maximising carbon savings throughout the duration of the project. Overall the team saved 62 tonnes of CO_2 (from 125 tonnes of CO_2 to 63 tonnes of CO_2).

The ideal solution was to reuse the existing sub-structure, replace the old deck with a new one and construct a new vehicle restrain parapet. The original bridge deck, constructed from large, longitudinal concrete slabs, was removed to install a new deck made from a steel frame and comparatively smaller pre-cast concrete slabs. The benefit of the latter was the requirement for a smaller crane for lifting and positioning, which reduced the carbon emissions from 48 tonnes of CO_2 to eight tonnes of CO_2.

The old deck material was crushed on site then reused to reinstate the car park, saving two tonnes of CO_2 by avoiding waste off site and import of aggregates.

The design of the vehicle parapet height was reduced from 1.5 m to 1 m. This was mainly for aesthetic purposes but it saved two tonnes of CO_2 from the amount of steel required and on its transportation.



Instead of using a sheet pile wall to separate the bank from the river the existing scour pile in the right-bank of the bridge was reused, instead of removing it, saving 18 tonnes of CO_2 in materials and transport.

The learning curve. The Sandford access bridge case study provides an example of a project seeking to maximise the reuse of existing site materials, which has led to a reduction in embodied CO_2 emissions, as well as making wider environmental gains and cost savings.

(Reproduced with permission from the Environment Agency – Case study IEM/2012/002.)

Weybridge 24-hour moorings

Background. Weybridge 24-hour moorings in Surrey was a £375,000 project to build a 120 m long river level footpath with access to moorings. The original design was based on the construction of an in-situ concrete wall supported by steel sheet piling. However, a design review was undertaken (primarily due to cost constraints and to meet a completion date).

The design was changed to utilise less costly and (overall) lower carbon materials whilst maintaining the same operational life.

Reducing the carbon footprint. The post construction emission was reduced by 169 tonnes of CO_2 compared with the original design (from 255 tonnes of CO_2 to 86 tonnes of CO_2).

Mesh filled concrete. The most significant CO_2 saving was from the 75% reduction in concrete used in the wall. A mesh filled with concrete was used instead of precast concrete blocks. This saved 173 tonnes of CO_2 (59 tonnes of CO_2 compared to 232 tonnes of CO_2 for precast concrete blocks). Originally a cast in-situ concrete wall was planned, which involved considerably more material.

Use of plastic piles. Further carbon savings were achieved through the innovative use of plastic piles (89.5% recycled) instead of steel sheet piles. The carbon footprint of the plastic piling was determined to be 8.6 tonnes of CO_2 compared to 17 tonnes of CO_2 for steel sheet piles.

Weybridge 24-hour moorings (continued)

Form liner. Using a dense foam form liner saved considerable time and cost compared to an authentic alternative. A brickwork finish was achieved to fit the surroundings using a reusable rubber form liner. This added four tonnes of CO_2 to the project.

Concrete specification. The concrete specification/grade was changed from exposure class XC3 to XC4 to increase the speed of the construction to meet a completion deadline. The carbon footprint increased by 27 tonnes, from 228 tonnes to 255 tonnes of CO_2 due to the time constraints.

The learning curve. The project was delivered on time and under budget. Overall the project team saved approximately £40,000 compared to the original design, with the same design life. Although the drivers for the fresh look at the design and consideration of innovative materials and approaches were predominantly cost and time, the carbon footprint reduced overall by 50%. This shows that innovation can have multiple benefits. It is well worth having a fresh look even without constraints being imposed.

(Reproduced with permission from the Environment Agency – Case study IEM/2012/001.)

Nottingham left bank

Background. Nottingham left bank is a £51 million scheme that will protect 16,000 homes and businesses from flooding. The project spans a 27 km reach of the River Trent and comprises flood defences with a mixture of earth bunds, concrete walls and sheet piling.

Due to constraints from working next to a railway and nature reserve, and the requirement to prevent groundwater seepage through the underlying sands and gravels, the project team came up with an innovative method that also lowered the total project carbon footprint.

Reducing the carbon footprint. Using the *Construction carbon calculator,* the total carbon footprint of the Nottingham project was 11,800 tonnes. The most significant contributions were from steel, concrete and material transport. The project team implemented many innovative changes to reduce the carbon footprint of the project. One of these examples resulted in changing the material and method of construction, which saved approximately 2,500 tonnes of CO_2 during this section of the works.

Use of TrenchMix. Sheet piles are traditionally used to form a cut-off to protect against groundwater seepage through defences, but on this project a large section of piling was replaced by the use of TrenchMix.

This process involves the mixing of the soil with cement; this cementitious material then sets, forming a barrier with a greatly reduced permeability. The soil and cement are mixed by a modified drainage trenching machine (*see photo*).

By using this method the team reduced the amount of steel used on the project by two-thirds, saving around 1,876 tonnes of CO_2.

Another benefit of the reduction in steel is the reduced carbon cost of transportation, saving around 700 tonnes of CO_2. The raw materials (cement) and plant emissions from the operation have a footprint of 330 tonnes of CO_2.

The learning curve. Even when taking into account the footprint of the materials and emissions from the TrenchMix method, it is still carbon beneficial by about 70% compared to the sheet pile option. Other advantages include less noise and vibration than sheet piling. This could be useful when working near sensitive areas, such as nature reserves and residential properties.

(Reproduced with permission from the Environment Agency – Case study IEM01/2011/002.)

Burrowbridge bank

Background. The Burrowbridge bank is a £270,000 project to repair a failing asset. This bank is part of the Parrett tidal reach and had suffered erosion, lost many of its timber piles that provided protection and had been affected by landslides. The project trimmed the riverside slope to a more suitable profile while maintaining crest width by filling at the rear. The embankment's toe was also protected by new timber piles and stone while a soft engineering solution was used for the bank slope.

Reducing the carbon footprint. Using the *Construction carbon calculator,* the predicted total carbon footprint of the Burrowbridge project was 140 tonnes. The most significant contributions are from import of clay and timber materials to site and export of trimmed arisings off site.

Reuse of material. The project team challenged the specification for re-profiling of the rear slope of the bund. This meant that material trimmed from the riverside slope could be reused on the land side of the bank. This saved having to import 550 tonnes of clay, reducing the number of lorry movements by 30 and saving around 15 tonnes of CO_2. It also reduced the transport of similar volumes of arisings off site, while providing cost savings of £5,500.

Reducing material used. Access to the site was challenging. Instead of using a traditional stone access track, the team hired in a temporary trackway, which allowed the team to go past the Burrow Mump (scheduled ancient monument) with minimum disruption. Using this trackway saved the need to import 1,500 tonnes of quarried material, which saved 20 tonnes of CO_2 and reduced an additional 100 lorry movements.

Recycled materials. Recycled hardwood timber piles were sourced from a local wharf being demolished, which saved approximately £35,000 and 26 tonnes of CO_2 (compared to using virgin hardwood).

The learning curve. This relatively small project reduced its total footprint by 60 tonnes of CO_2 – a reduction of over 40% of the predicted footprint. This project team proved that by looking at material selection and challenging the specification, large savings can be made not only in carbon and project costs.

(Reproduced with permission from the Environment Agency.)

Radcot weir

Background. The Radcot weir project is one of five sites in the Paddle and Rymer package. The main construction work at Radcot involves removing the existing Paddle and Rymer structure and replacing it with a dipping radial gated weir. This will benefit operators of the weir through the removal of health and safety risk, easier operation and standardisation of the weir. In addition the project will provide a new bypass channel to allow upstream and downstream movement of fish and provide recreational use for canoeists.

Using the *Construction carbon calculator*, the total carbon footprint of the Radcot weir project is 600 tonnes. The most significant contributions are from concrete and steel. The project team, by looking at material selection, have saved around 50 tonnes of CO_2 during the first phase of works.

Reducing the carbon footprint. Use of granular ground blast furnace slag (ggbs) in concrete; 50% ggbs replacement was used in the base and 70% in the rest of the structure. This saved around 40 tonnes of CO_2 (a saving of over 60% when compared to a CEM1 concrete).

Reuse of material. The old structure was demolished and then crushed on site. This material was then used underneath the blinding instead of a primary aggregate, saving both the cost and carbon of disposal of the material and the import of primary aggregate. This saved around 5.2 tonnes of CO_2, resulted in 40 less lorry movements and saved £10,000.

Avoiding waste. The team cast the coping stones for the structure on site by using the leftover concrete from the back end of the concrete pours, which would normally be wasted. The coping stones were produced over time meaning that no extra concrete was ordered for their production, saving around 1.2 tonnes of CO_2 and around £2,000 to £3,000.

Radcot weir (continued)

Additional benefits. The big win from the project was the use of GGBS as a replacement in the concrete. This also has many other benefits, apart from the saving in CO_2, including lighter colour (desirable in this case), lower early thermal cracking, higher strength development over time, increased durability and increased workability. A potential disadvantage was noted as slower, early strength development/longer striking time, which can increase the construction programme, hence only 50% replacement being used in the base, but in this case the concrete typically achieved a strength of 30 N/mm^2 at seven days.

These reductions are from only half of the project; the second phase, which was due to start later, is estimated to save at least an equivalent amount resulting in a final saving of around 100 tonnes, which is almost 20% of the estimated project footprint, as well as potentially £25,000 in costs. For future projects, the team is looking at saving further CO_2 by potentially reducing the criteria for crack control steel reinforcement.

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Archaeology and heritage

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Supporting **INFORMATION**

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.

Scan the QR code to complete a brief survey, and share your feedback on the course.

Summary of archaeology and heritage legislation and guidance

This list is not exhaustive and only includes legislation mentioned in this section of GE700.

Legislation and guidance	Enforcement agencies*				
	CADW	HE	HES	LA	NIEA
Acts (primary legislation)					
Ancient Monuments and Archaeological Areas Act 1979	✓	✓	✓		
Burial Act 1857	✓	✓	✓		✓
Disused Burial Grounds Act 1884	✓	✓			
Planning (Listed Buildings and Conservation Areas) Act 1990	✓	✓			
Planning (Listed Buildings and Conservation Areas) (Scotland) Act 1997			✓		
Protection of Military Remains Act 1986	✓	✓	✓		✓
Protection of Wrecks Act 1973	✓	✓	✓		✓
Regulations (secondary legislation)					
Listed Building Consent				✓	
Planning (Listed Buildings and Conservation Areas) (Heritage Partnership Agreements) Regulations 2014		✓			
Scheduled Monument Consent	✓	✓	✓		✓
Town and Country Planning (Historic Environment Scotland) Amendment Regulations 2015			✓		
Guidance					
Conserving and enhancing the historic environment	Ministry for Housing, Communities and Local Government				
National planning policy framework					
Net Regs Environmental guidance					
Organisations providing information and guidance on their websites	✓	✓	✓	✓	✓

***Key**

CADW	The Welsh Government's historic environment service
HE	Historic England
HES	Historic Environment Scotland
LA	Local Authorities
NIEA	Northern Ireland Environment Agency

Overview

The heritage environment is a precious resource bringing millions of tourists to the UK to enjoy our rich, cultural past. Archaeology is a major factor in construction and development and an estimated £218m is spent by developers on archaeology in the UK each year. Heritage is important, as it records the past and provides a sense of place.

This chapter gives a general overview of the legal framework for the protection and management of historic buildings, monuments, archaeological sites and remains. It identifies what constitutes the historic environment, how this environment's features are protected in law and where to obtain consent to carry out works on or near to them.

The chapter also gives guidance on the measures to be taken to protect historic and archaeological features during the works and the actions to take if an unexpected archaeological discovery is made.

4.1 Introduction

Archaeology is often underestimated as a business or project risk so is often not considered early enough in the feasibility or design stages of projects, leading to unexpected or unplanned consequences.

Archaeological remains and the historic environment provide a valuable and irreplaceable part of a nation's history and identity. The use of local and traditional materials in buildings creates a sense of place and attachment for the population. For these reasons archaeology and built heritage form an important element of planning policy and must be considered early in any construction project.

Today, archaeology is a significant element of construction and development, and all parties (developers, archaeologists, conservation officers and the regulatory authorities) should aim to ensure that good practice is applied.

The main reasons why the construction sector should address archaeology and the historic environment are shown below.

Planning law, heritage law and planning policy. Archaeological remains and historic buildings are an important part of our cultural heritage. They are a fragile, finite and irreplaceable resource that needs to be protected. This is recognised within the UK planning process and by national legislation and guidance. Whilst some archaeological sites are protected by law, all archaeological remains affected by development are treated as material considerations in the planning process. Local Authorities should maintain historic environment records and use these as a database to inform planning applications.

Education and new knowledge. Archaeology and the historic environment make a significant contribution to education, social cohesion and the economy and bring wide public benefits and knowledge through archaeological and historical education in schools and universities.

Economy and society. Archaeology, historic buildings and landscapes underpin the UK tourism and heritage industries and create jobs. They also play an important role in regeneration by invoking a sense of place, belonging and cultural identity for new developments.

Sustainability. Developers can demonstrate their commitment to sustainability and responsible development through a proactive approach to archaeology and the historic environment, and also a recognition that archaeological remains are a non-renewable resource that need environmental protection and management.

An early, positive and proactive approach to archaeological remains and historic buildings can:

- minimise the impacts
- provide early effective risk management in the planning phase, minimising the risk of unexpected archaeological finds, which would have significantly higher costs attached if found during the construction phase
- heighten a sense of place for new developments
- provide opportunities for involving the community, leading to positive publicity for the construction project.

4.2 Important points

A heritage statement (as part of an overall environmental statement or environmental impact assessment) may be required to support a planning application and may also require some intrusive works on the site. The heritage statement will identify the significance of heritage assets and the effects the proposed development will have on them.

The level of detail required within the statement will depend upon the significance of the asset (for example, a Grade 1 building will require more detail than Grade 2). Additional planning requirements will include a heritage management plan to identify the controls that are required during the works.

A professional archaeologist may be required, as a condition of any planning consent, to supervise the works as they proceed and to ensure that suitable protection measures are put in place and maintained.

An archaeologist may also be required to excavate and record remains that are due to be removed.

Suitable controls to manage archaeology and historic sites should be included in the project as early as possible for the following reasons.

- To comply with legal requirements relating to scheduled monuments, listed buildings and protection of the historic environment.
- To ensure buildings are designed to avoid the disturbance of remains and to preserve historic features.
- To enable designers to incorporate historic features of the site in the final development.
- To avoid disturbance during the construction process itself by the following means.
  - Ensuring all archaeological works are segregated from the main works with authorised entry.
  - Identifying existing underground services that might have an impact on archaeology.
  - Ensuring archaeological excavations are suitably protected during the works.
  - Avoiding dewatering work in the vicinity of archaeological remains.
- For guidance for working on projects in the UK set against or within the heritage environment, the following organisations should be contacted.
  - Historic England.
  - Department for Communities in Northern Ireland.
  - Historic Environment Scotland.
  - Cadw in Wales.

4.3 Protected monuments, buildings and sites

Heritage assets range from sites and buildings of local historic value to World Heritage sites that have the highest significance of international value.

Each asset is an irreplaceable resource and should be conserved so that they can be enjoyed by existing and future generations.

The main guidance in the UK is found in the following documents.

- National Planning Policy Framework (NPPF) (England).
- Strategic Planning Policy Statement for Northern Ireland.
- Scottish Planning Policy and the National Planning Framework.
- Planning Policy Wales.

www **Historic England has *Good practice advice in planning* (Note 3) and *The setting of heritage assets* available on its website.**

Guide to the conservation of historic buildings (BS 7913) provides information on the principles of conservation and restoration of historic buildings for architects, designers, surveyors and contractors.

It outlines the importance of using appropriate materials for maintenance and the principles behind honest repairs (work that is clearly identifiable as being a repair and is not disguised to look like part of the original building or structure).

It also explains how traditional buildings perform differently to modern buildings and highlights the damaging effects that the retrofitting of energy efficiency measures can have on a building.

www **For further guidance on planning in England, Northern Ireland, Scotland and Wales visit the relevant website.**

4.3.1 UNESCO World Heritage sites

World Heritage site status is the most significant form of protection. It is awarded by the United Nations Educational, Scientific and Cultural Organisation (UNESCO) to places or buildings that have been judged to be of outstanding universal value and are the best examples of the world's culture and/or natural heritage.

Some examples from the UK are given below.

- Stonehenge, an archaeological site in England.
- Giant's Causeway, natural environment in Northern Ireland.
- Edinburgh, a cultural city in Scotland.
- Blaenavon, an industrial landscape in Wales.

Stonehenge is a well known scheduled ancient monument

4.3.2 Scheduled monuments

The word *monument* covers a range of archaeological sites. The term *scheduled ancient monuments* (SAMs) is often used (however, not all scheduled monuments are ancient). Further, scheduled monuments are not always visible above ground or well known (for example, wells and wartime gun emplacements).

Nationally important sites, buildings and monuments are given legal protection by being placed on a schedule for monuments. The UK heritage bodies identify sites that should be placed on the schedule by the respective government.

The Ancient Monuments and Archaeological Areas Act 1979 supports a formal system of scheduled monument consents for any works to a designated monument.

All work to a monument that is scheduled as an ancient monument requires scheduled monument consent, as do any works to the grounds that surround it.

Some scheduled monuments are not visible in the landscape, such as this medieval well under the Doncaster market

The following are criminal offences, potentially leading to fines or imprisonment.

- Carrying out unauthorised works on a scheduled monument without consent.
- Causing damage to a scheduled monument.
- Failing to adhere to the terms of a consent.

Development or construction works that affect a scheduled monument require formal permission (scheduled monument consent).

The granting of scheduled monument consent does not imply planning permission, or vice versa.

Applications for scheduled monument consent are made:

- in England, to the Department of Culture, Media and Sport, through Historic England
- in Northern Ireland, to the Department for Communities
- in Scotland, to Historic Environment Scotland
- in Wales, to Cadw.

4.3.3 Listed buildings

A listed building has statutory protection against unauthorised demolition, alteration and extension.

The grades of listed building are shown in the table below (Grades I and A/Category A being the highest level of protection).

England and Wales	Scotland	Northern Ireland
Grade I is the highest grade, reserved for buildings of exceptional interest	**Category A** buildings are of 'national or international architectural or historic importance'	**Grade A** buildings display excellent architecture of a particular period or style
Grade II* buildings are special interest, and carry more significance than Grade II alone	**Category B** buildings have a regional significance	**Grade B+** are high quality buildings, but fall short of Grade A due to an impairment
Grade II listed buildings account for over 90% of all listed buildings	**Category C** buildings are of local importance, and are usually simple buildings	**Grade B1** buildings are a good example of a particular period or style
		Grade B2 buildings qualify by virtue of only a few attributes

Any works to be undertaken to a listed building that would affect its character, or the character of any building or structure in its curtilage (area of land attached to a building), require a listed building consent.

It is a criminal offence to demolish, alter or extend a listed building without consent. Where work has been carried out without consent, and the character and appearance of the building are affected, a listed building enforcement notice may be served. This will require reinstatement of the building back to its original condition or in keeping with the building's criteria of listing prior to works being carried out. If this is not feasible a fine, sentence or both may be served.

It should be noted that Planning Authorities and National Park Authorities have the power to serve a building preservation notice on the owner of a building that is not listed, but which they consider is of special architectural or historic interest. The building is then protected for six months, like a listed building, in which time the Secretary of State must decide whether to list the building.

4.3.4 Conservation areas

A *conservation area* is defined as an area that has special architectural or historic character that is desirable to preserve or enhance. Every Local Authority in England has at least one conservation area, and there are over 11,000 across the UK.

Local Planning Authorities are responsible for conservation area designation and policy, both of which seek to protect the character of the whole area, including open spaces, not just individual buildings.

Conservation areas are given greater protection by the Local Authority, which has extra control over demolition, minor developments and the protection of trees. In the past, conservation area consent from the local Planning Authority was required prior to works being carried out in the designated area.

The requirement for conservation area consent, however, was abolished in the Enterprise and Regulatory Reform Act and replaced with a requirement for planning permission for demolition of, or alterations to, a building in a conservation area.

4.3.5 Parks and gardens

Historic parks and gardens can be registered, which means inclusion on the register of parks and gardens of special historic interest. Alterations to parks and gardens generally do not require statutory consent unless a planning application is required or it affects a tree covered by a tree preservation order.

It should be noted that local Planning Authorities are encouraged to include policies in their development plans for the protection of registered historic parks and gardens.

4.3.6 Designated wrecks

Section 1 of the Protection of Wrecks Act 1973 provides protection to some 60 wrecks around the UK, designated for their archaeological and historical significance. In Northern Ireland these are regulated under the Historic Monuments and Archaeological Objects (Northern Ireland) Order 1995.

The administration of the Act and the issue of licences allowing diving, survey, collection of objects or excavation on these sites is the responsibility of the following organisations.

- Historic England.
- Department for Communities in Northern Ireland.
- Historic Environment Scotland.
- Cadw in Wales.

4.3.7 Designated vessels and controlled sites

Certain military aircraft crash sites and military shipwrecks are designated and protected under the Protection of Military Remains Act 1986. These sites are administered by the Ministry of Defence.

4.3.8 Burial grounds and human remains

In England and Wales, authorisation to disturb human remains may be granted by Act of Parliament, Home Office licence (through the Ministry of Justice) or Church of England Faculty. Under Scottish law there is no such system.

In England, disturbance of human remains may be allowed by an Act of Parliament that authorises a specific major project, but otherwise the following legislation applies.

- Town and Country Planning (Churches, Places of Religious Worship and Burial Grounds) Regulations 1950.
- Disused Burial Grounds Act 1884.
- Disused Burial Grounds (Amendment) Act 1981.
- Burial Act 1857.

If any form of construction work is to take place on a disused burial ground, then the Disused Burial Grounds (Amendment) Act stipulates that human remains should be removed before work begins. However, if planned works will leave human remains undisturbed, then dispensation can be obtained from the Home Office authorising that the burials remain in situ.

4.3.9 Treasure

Under the Treasure Act 1996 (England, Wales and Northern Ireland), certain finds are deemed as treasure. These must be reported to the coroner for the district within 14 days, who will advise on the course of action to be taken. Under the Act, the following finds would be classed as treasure:

- Coins that are at least 300 years old (coins are usually only treasure if there are 10 or more, or if they are with other items that meet the treasure definition).
- Objects that contain at least 10% of gold or silver and are at least 300 years old.
- Any object that is found in the near vicinity of known treasure.
- Base metal deposits of prehistoric date.

In Scotland any treasure finds are controlled by a system called Treasure Trove. The role of Treasure Trove is to ensure that objects of cultural significance from Scotland's past are protected for the benefit of the nation and preserved in museums across the country.

At the time of publication, the definition of treasure was being consulted upon in Parliament.

Proposed changes will help protect thousands more objects of exceptional archaeological, historical and cultural importance. This will ensure that more new discoveries can go on public display in museums across the country for the public to see and enjoy in England, Wales and Northern Ireland.

The proposed new criteria will apply to the most exceptional finds over 200 years old – regardless of the type of metal of which they are made, so long as they provide an important insight into the country's heritage. This includes rare objects, those which provide a special insight into a particular person or event, or those which can shed new light on important regional histories. Discoveries of treasure meeting these new criteria will be assessed by a coroner, and will go through a formal process in which they can be acquired by a museum and go on display to the public.

The statutory instrument to widen the definition was laid before Parliament on Monday 20 February 2023. If Parliament approves the change, it will come into force four months after signing.

4.4 Managing archaeology

In project terms, archaeological remains should be considered as risks which, if not properly identified and mitigated, may cause an adverse effect on a development.

Typical archaeological risks faced by development include:

- planning constraints and potential for spot listing and scheduling of remains
- unforeseen or unexpected archaeological remains affecting design, implementation programme and profit margins.

The environmental statement and planning conditions for the project, provided by the client, should identify any obligations for the management and protection of archaeology and historic buildings, monuments or sites. These requirements should be incorporated into the project environmental management plan (EMP) in a sub-section for heritage management. In some cases it will be a condition of the planning consent to prepare an archaeological management plan.

Where this is the case, it is good practice to employ a qualified archaeologist to plan, recommend actions and identify areas of responsibility in relation to the following.

- What planning policy and legislation authorities will consider when deciding how to treat any archaeological remains.
- The types of archaeological site and range of features that may occur.
- The range of non-intrusive and intrusive techniques that archaeologists will use to assess and evaluate archaeological remains.
- The main roles of the client, archaeological consultant, contractor and curator.

It is often best to deal with archaeological issues (including excavation) in advance of the main construction phase.

All field evaluation work should be completed and an agreed mitigation strategy (if required) should be in place before construction or groundwork starts.

It will also be important to agree the appropriate method of working adjacent to sensitive areas as vibration from operations (such as excavation or tunnelling) may cause damage.

There may be an obligation to provide vibration monitoring of the works to ensure that vibration levels are not exceeded.

It may also be a requirement of the planning consent to employ a qualified archaeologist to carry out a watching brief during construction works (such as for excavations or dismantling works). All areas of known archaeological or historic interest should be protected with suitable fencing to prevent damage or encroachment, with access to site traffic and personnel carefully controlled. This is a legal requirement for scheduled sites.

In respect to health and safety, any excavations for archaeological works should be treated as usual and protected to avoid inadvertent access by personnel or site equipment. The location of underground services should also be identified, as these could cross sites where archaeological excavation is planned. Consideration should be given to the possibility of the presence of materials that can be harmful to health (such as lead paints, asbestos-containing materials and horse-hair plaster, which may contain anthrax spores).

Many archaeological excavations will also take place on brownfield sites that have been contaminated. If there is reason to believe that the ground is contaminated, arrangements should be made to undertake sampling and testing before archaeological work starts on site.

Dewatering schemes can have a negative impact on archaeological features as they could cause differential settlement or damage to materials that have previously been protected by being waterlogged. Appropriate methods of dewatering should be agreed in advance of the works taking place.

4.5 Unexpected discovery

If suspected archaeological objects or remains are found during the construction works without an archaeologist on site, certain procedures should be followed. The archaeological finds should be protected and not be disturbed any further before any specialist investigation and advice has been obtained.

Advice should be sought by contacting the Local Authority archaeological officer on how to proceed.

Accidental discovery

In the case of accidental discovery of archaeological finds or human remains, the following sequence of actions are considered good practice.

- Stop work in the area of the discovery.
- Leave the find in situ and undisturbed.
- Control access to the area to authorised persons only.
- Stop vehicle traffic entering the area.
- Report the find to the site manager.
- Take specialist advice as appropriate.
- Report human remains, treasure and other archaeological finds to the appropriate statutory authority (usually the coroner).
- Report the find to the appropriate local planning authority or other curator.

Under the Burials Act there is a requirement to report unexpected discoveries of human remains to a coroner.

If there is any possibility that the remains are recent (for example, less than 100 years old) the local police should be contacted to determine whether the remains are ancient or not.

For authorisation to continue, approval will need to be given by the Home Office.

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Legislation and guidance	Enforcement agencies*				
	LA	NE	NIEA	NRW	NS
Acts (primary legislation)					
Amenity Lands Act (Northern Ireland) 1965			✓		
Conservation (Natural Habitats etc.) (Amendment) Regulations (Northern Ireland) 1995			✓		
Countryside and Rights of Way Act 2000		✓	✓	✓	✓
Countryside (Scotland) Act 1967					✓
Environmental Protection Act 1990		✓		✓	✓
National Parks and Access to the Countryside Act 1949	✓	✓	✓	✓	✓
Pesticides Act 1988	✓	✓	✓	✓	✓
Planning (Northern Ireland) Order 1991			✓		
Protection of Badgers Act 1992		✓		✓	✓
Town and Country Planning Act 1990	✓	✓		✓	
Town and Country Planning (Scotland) Act 1997					✓
Tree Preservation Orders	✓				
Weeds Act 1959		✓		✓	✓
Wildlife and Countryside Act 1981		✓		✓	
Wildlife and Countryside (Amendment) Act 1991 [England, Scotland and Wales]		✓		✓	✓
Wildlife (Amendment) (Northern Ireland) Order 1995			✓		
Wildlife and Natural Environment Act (Northern Ireland) 2011			✓		
Regulations (secondary legislation)					
Conservation of Habitats and Species Regulations 2017	✓	✓	✓	✓	✓
Control of Pesticides Regulations 1986		✓	✓	✓	✓
Environmental Permitting (England and Wales) Regulations 2016		✓		✓	
Hazardous Waste (England and Wales) Regulations 2005		✓		✓	
Hedgerows Regulations 1997 [England and Wales only]	✓				
Town and Country Planning (Environmental Impact Assessment) Regulations 2017		✓	✓	✓	✓
Guidance					
CIRIA publication *Working with wildlife: guidance for the construction industry*					
INNSA Code of Practice *Managing Japanese knotweed*					
Net Regs Environmental guidance					
Organisations providing information and guidance on their websites	✓	✓	✓	✓	✓

***Key**

CIRIA	Construction Research and Information Association
INNSA	Invasive Non-Native Specialists Association
LA	Local Authorities
NE	Natural England
NIEA	Northern Ireland Environment Agency
NRW	Natural Resources Wales
NS	NatureScot